

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/382467665>

A simple hydrodynamic model for clay

Article in *Journal of the Mechanics and Physics of Solids* · July 2024

DOI: 10.1016/j.jmps.2024.105789

CITATIONS

4

READS

289

2 authors:



Max Wiebicke

The University of Sydney

18 PUBLICATIONS 512 CITATIONS

SEE PROFILE



Itai Einav

The University of Sydney

204 PUBLICATIONS 6,000 CITATIONS

SEE PROFILE



A simple hydrodynamic model for clay

Max Wiebicke, Itai Einav^{*}

Sydney Centre in Geomechanics and Mining Materials, School of Civil Engineering, The University of Sydney, 2006, Sydney, Australia

ARTICLE INFO

Dataset link: github.com/MaxWiebicke/Terracotta-pq

Keywords:

Clay
Thermodynamics
Rate-dependent phenomena
Solid mechanics

ABSTRACT

Laboratory description of clay normally distinguishes the scale of atoms from the scale of clay particles and aggregates. Contemporary constitutive models for clay tend to ignore this scale separation, and rather focus on phenomenology. By considering scale separation, this paper introduces a robust physics-based phenomenological constitutive model for clay that qualitatively captures their broad spectrum of rate-dependent mechanical features. The model is derived using the thoroughly rigorous hydrodynamic procedure. While some imagine that by considering rigour and physics, their models would get complicated, the resulting set of equations reveal a surprising degree of simplicity. The derivation strongly benefits from the principle of two-stage irreversibility, which describes energy flow within the material from the continuum scale down to the atomistic micro-scale, through the meso-scale of clay aggregates. While thermal and meso-related temperatures capture atomistic and clay aggregate fluctuating motions, a sink term from the latter to the former underpins the direction of the energy flow. The model's standout feature is in pinpointing new transport coefficients that drive both volumetric and shear plastic flows in a thermodynamically coupled manner. A novel scheme is then proposed to calibrate these coefficients from conventional steady-state observations. Thanks to the formulation the model shows a remarkable level of predictiveness, despite being relatively simple mathematically. In particular, the model readily explains the broad spectrum of rate-dependent phenomena during transient loading, along with creep and relaxation processes. Given the generality of hydrodynamics, it is anticipated that the new model could be expanded to capture fluid-solid transitions between liquid-like soft mud and plastic-like stiff clay responses, contingent on water content variations.

1. Introduction

Clay is a fine-grained material that is found abundantly on Earth and exhibits a complex stress–strain response when subjected to external loading. Understanding and accurately modelling this response is of fundamental significance across multiple scientific disciplines. Over the years, many constitutive models have been developed for clay, with the most predictive ones typically based on fiddly mathematical frameworks, such as elastoplasticity, which require specialised expertise for implementation. The intricacy of those formulations has limited their accessibility to the broader audience of engineers (Kolymbas, 2000), and often causes numerical problems when applied to solving boundary value problems. On the other hand, we demonstrate that a rigorous and predictive model may be as simple as the ‘Terracotta’ model¹ proposed in this paper, and depicted by Fig. 1. The first aim of this paper is to present the fundamental physics and careful derivation that led us to arrive at such a relatively simple mathematical structure. The second aim

^{*} Corresponding author.

E-mail address: itai.einav@sydney.edu.au (I. Einav).

¹ Over six decades ago, a renowned clay model was named after river cam by its developers who worked nearby. Here, we follow this path with ‘Terracotta,’ paying tribute to the Terra Cotta Roasters Cafe near the University of Sydney, where daily coffees inspired its development.

$$\begin{array}{l}
\text{stresses} \left\{ \begin{array}{l} p = \frac{1}{2} \phi^6 \left(\tilde{K} \varepsilon_v^e{}^2 + 3\tilde{G} \varepsilon_s^e{}^2 \right) + \overbrace{2\frac{\alpha}{\Gamma} T_m \dot{\varepsilon}_v}^{p^d} + \overbrace{\frac{1}{\Gamma} T_m^2}^{p_m^T} \\ q = \underbrace{3\tilde{G}\phi^6 \varepsilon_v^e \varepsilon_s^e}_{q^e} + \underbrace{2\frac{\gamma}{\Gamma} T_m \dot{\varepsilon}_s}_{q^d} \end{array} \right. \\
\\
\text{rate of states} \left\{ \begin{array}{l} \dot{T}_m = \alpha \dot{\varepsilon}_v^2 + \gamma \dot{\varepsilon}_s^2 - \eta T_m^2 \\ \dot{\phi} = \phi \dot{\varepsilon}_v \\ \begin{bmatrix} \dot{\varepsilon}_v^e \\ \dot{\varepsilon}_s^e \end{bmatrix} = \begin{bmatrix} \dot{\varepsilon}_v \\ \dot{\varepsilon}_s \end{bmatrix} - \underbrace{\frac{T_m}{p_1 \phi^\lambda} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} p^e \\ q^e \end{bmatrix}}_{\dot{\varepsilon}^p} \end{array} \right. \\
\\
\text{transport coefficients} \left\{ \begin{array}{l} a = \sqrt{\frac{\eta}{\alpha}} \\ b = -a \frac{q^e}{M^2 p^e} \\ c = \sqrt{\frac{\eta}{\gamma}} \frac{1}{M \omega} + \frac{a}{M^2} \end{array} \right.
\end{array}$$

Fig. 1. The ‘Terracotta’ model – all the required equations. In the above, p and q denote the volumetric and deviatoric stress invariants; ε_v and ε_s the volumetric and deviatoric strain invariants; ϕ the solid fraction; T_m the meso-related temperature; superscripts e , d , T , p refer to the elastic, dissipative (or viscous), thermodynamic and plastic contributions; $\tilde{K}$, $\tilde{G}$, M , ω , p_1 , λ are material constants; α , γ , η are coefficients, chosen to be constant in this paper, while $\Gamma = 1 \text{ K}^2/\text{kPa}$ is here used as a fixed coefficient that sets the unit of the meso-related temperature.

is to demonstrate the great capacity of the model to predict an impressive wide range of rate-dependent experimental observations, despite being so simple mathematically.

In order to arrive at a meaningful model, it is important to briefly discuss the structure of clay at various scales and its consequences. At the micro-scale, like any other material the representative volume of clay contains an overwhelming amount of atoms and molecules, often in the form of minerals sheets (Grim, 1962). These micro-structures tend to stack and form mesoscopic platy particles (platelets), which further agglomerate into aggregates depending on the environmental conditions. Due to the chemistry associated with the reaction and deformation of platelets and aggregates, clays show a rich and complex rate-dependent mechanical behaviour (Mitchell and Soga, 2005; Le et al., 2012). Some aspects of the rate-dependency are featured in complex stress–strain responses during changing loading rates (Leroueil et al., 1985; Tatsuoaka et al., 2002; Sorensen et al., 2007), creep (Buisman, 1936) and relaxation testing protocols (Lacerda and Houston, 1973).

In trying to capture those rate-dependent phenomena, constitutive models have been predominantly formulated within the frameworks of elastoplasticity (Adachi and Oka, 1982; Borja and Kavazanjian, 1985; Yin and Graham, 1999; Yuan and Whittle, 2021) and hypoplasticity (Niemunis and Krieg, 1996; Niemunis, 2003; Fuentes et al., 2018). A comprehensive review of the typical previous approaches to include rate-dependency in these frameworks can be found in Tafil et al. (2020). The majority of these approaches relies on the ‘isotach’ concept (Šuklje, 1969), which assumes that the viscous or visco-plastic strain rate solely depends on the current state of stress and density. However, experiments on clay with step-changes of the strain rate during shear or compression sometimes reveal responses which deviate from the isotach concept (Sorensen et al., 2007; Tatsuoaka et al., 2008). Furthermore, it has been demonstrated, that such models predict too much creep at the field scale when calibrated using laboratory scale experiments (Yuan and Whittle, 2013). Recently, Yuan and Whittle (2018) proposed a 1D elasto-viscoplastic formulation that is able to capture both isotach and non-isotach responses to changes in the loading rate by introducing a new state variable that captures the strain rate history. This formulation was further implemented in their more general and mathematically intricate bounding surface sort of model for clay (Yuan and Whittle, 2021).

In this work, we are seeking to develop a physics-based model that is both simple mathematically as well as capable of predicting the diverse range of rate-dependent behaviours observed in clay. To accomplish this goal, the model is constructed based on fundamental principles rather than adapting an existing model, thereby circumventing potential pitfalls associated with prior misconceptions. The derivation follows the hydrodynamic procedure, ensuring a thermodynamically sound structure that surpasses past formulations in soil mechanics. In particular, unlike previous thermodynamic models for clay, which merely consider the first and second laws, the derivation of Terracotta benefits from additional thermodynamic principles associated with past progress using the hydrodynamic procedure, including:

- the scale separation of temperature in heterogeneous media,
- the energy flow across the corresponding scales,

- the reciprocity of irreversible thermodynamic fluxes,
- the vanishment of non-equilibrium properties close to thermodynamic equilibrium, and
- the contribution of a thermodynamic pressure to the overall pressure.

The hydrodynamic procedure was originally formulated for general media by Landau and Lifshitz (1980, 1987) and was later extended to granular media by Jiang and Liu (2009), who called their formalism granular solid hydrodynamics (GSH). Constitutive models that are formulated within GSH have demonstrated the ability to describe a wide range of physical phenomena across different rate regimes that are commonly not captured by the classical elasto- or hypoplasticity formulations (Jiang and Liu, 2015). Nevertheless, despite introducing some well-defined concepts during their development, current GSH models are not practical due to an abundance of complex, imagined nonlinear relationships. These problems partly arise because those GSH models are based on an overwhelmingly large number of transport coefficients² (Onsager, 1931) without any clear scheme to determine them. The transport coefficients are pivotal for such models, as they come to describe non-equilibrium thermodynamic processes such as the plastic and viscous flow. As a result, GSH models for sand Jiang and Liu (2009, 2014) and clay (Zhang and Cheng, 2015, 2016) come with an excessive amount of parameters, which are very hard to determine.

To reduce the complexity of parameterisation and enhance predictive robustness, one approach was to integrate certain aspects of the hydrodynamics procedure with principles from plasticity theory (Alaei et al., 2021). While this integrated approach has shown promising results, including by elucidating intricate phenomena like dilation and breakage in sand (Alaei et al., 2022) and recurrent stress fluctuations in porous media (Riley et al., 2023b), it inherently demands navigating through the complex mathematical abstraction of plasticity, particularly in formulating the non-negative plasticity multiplier, whose symbolic expression may often span several pages.

In this paper, we present a simple model for clay using the original concepts of hydrodynamics and some of the ideas developed by Jiang and Liu (2009) for sand and later for non-equilibrium soil–water retention in soils by Einav and Liu (2023). The model is readily shown to capture major features of clay behaviour, such as critical state behaviour and rate effects while avoiding the use of physically abstract concepts such as bounding or yield surfaces and the non-negative multiplier in mathematical plasticity formulations. By identifying previously ignored transport coefficient for thermodynamically coupled volumetric and shear plastic flows, and devising a novel way to determine the transport coefficients, we eliminate the necessity of relying on abstract concepts from the mathematical theory of plasticity.

This paper gradually unfolds the physical and mathematical origin of all the quantities and parameters in Fig. 1, and then tests their validity. In order to achieve this, the paper is organised as follows. We start by briefly describing the concept of meso-related temperature as an essential state variable for capturing correctly rate-dependent phenomena in clay. Next, we briefly review the requirements stemming from the hydrodynamic procedure on the structure of models involving this meso-related temperature as a state variable, as well as the conventional thermal temperature, density and elastic strain. Given those requirements, we move on to derive all the ingredients of the Terracotta constitutive model displayed in Fig. 1. The ability of this model to predict stress–strain phenomena of clays is then assessed through comparisons against diverse experimental observations. Upon conclusion, we will highlight the perspectives of the newly proposed model in light of the generality of the hydrodynamics procedure and the special mathematical structure of models developed through it.

2. Meso-related structures and temperature in clay

In the following we propose the use of a meso-related temperature as a non-equilibrium state variable of clay. Following original ideas of Jiang and Liu (2009) for granular materials, the introduction of this variable helps separating the thermodynamic scale of mesoscopic structures from the thermodynamic scale of microscopic (atomistic and molecular) structures. The meso-related temperature T_m is introduced to account for the average kinetic energy of the velocity fluctuations of the meso-related structures, while the conventional thermal temperature T is used to capture the jiggling of atoms and molecules that are considered microscopic.

In the case of sand, the meso-related structures are commonly identified by the grains and T_m is referred as the granular temperature T_g (Jiang and Liu, 2009) that quantifies the average kinetic energy of velocity fluctuations (jiggling) of grains (Walton and Braun, 1986; Goldhirsch, 2008). The granular temperature may be connected to the kinetic pressure, which can be measured experimentally (Maranic et al., 2021).

In the case of clay, the basic building blocks are mineral sheets (Grim, 1962), whose thickness is of the order of several atoms. Clay particles are assemblies of such stacked sheets, often in the form of platelets, and can eventually aggregate to form larger clusters of particles as visualised in Fig. 2. Due to the small size of clay sheets, the electro-chemical interactions between them are complex and depend on environmental factors such as the acidity and the salinity among others (Santamarina et al., 2001; Mitchell and Soga, 2005). When a macroscopic clay sample is being perturbed by straining or stressing, the aggregates in it are imagined to fluctuate at that mesoscopic scale, just like grains in granular media.

Two typical and very different clay minerals are kaolinite and montmorillonite. The representative size of kaolinite aggregates observed in suspensions is larger than 1–2 μm and therefore at least five orders of magnitude larger than the atoms (Gupta et al., 2011; Mansa et al., 2017). Aggregates of montmorillonite are commonly at least 200 nm large (but more commonly larger) (Tombácz et al., 2001; Furukawa et al., 2009; Missana et al., 2018). These structures tend to be the smallest ones identified and measured

² The term ‘coefficients’ is distinguished from material parameters and represents a function of state. Those coefficients typically depend on parameters. On the other hand, ‘parameters’ and ‘constants’ would be used interchangeably to denote state-independent values of a given material.

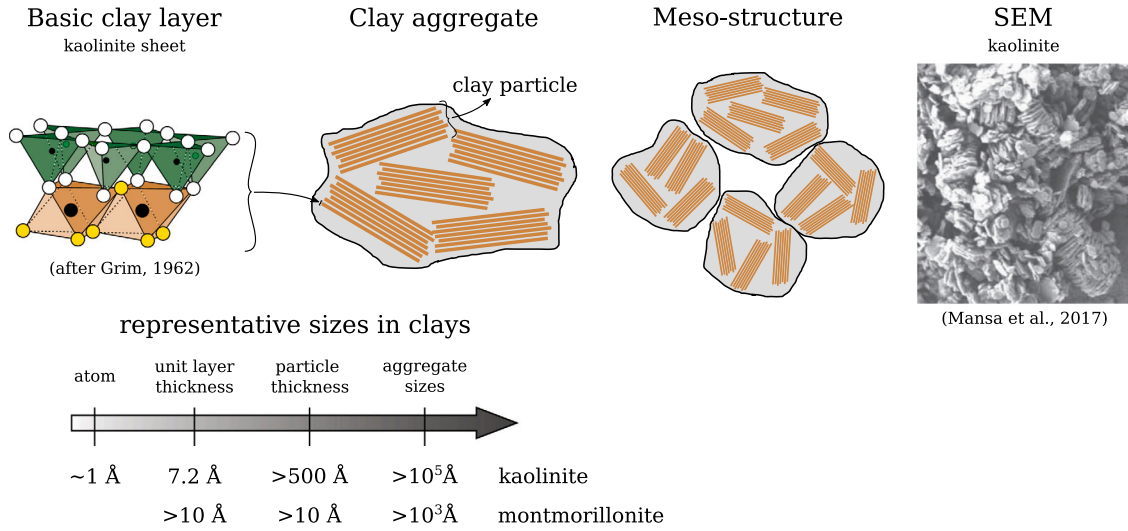


Fig. 2. Distinct structures across the scales of clay, here illustrated for kaolinite minerals.

in clay suspensions for which the natural structure has either been destroyed due to remoulding or the suspensions have been prepared from powders and the particles have just aggregated to form larger structures. Natural sizes of typical aggregates can thus be assumed to be even larger (Pysrillos et al., 1999; Tan et al., 2017). Based on this clear difference between the sizes of atoms and molecules and the sizes of clay particles and aggregates, we consider two separate scales associated with the internal motions that develop within representative samples of clay materials. The fluctuating collective motion of atoms and molecules is attributed to the micro-related thermal temperature, while the fluctuating collective motion of clay particles and aggregates is attributed to the meso-related temperature.

The concept of the meso-related temperature in clay is illustrated in Fig. 3. When clay is deformed, large pores between aggregates and individual particles collapse or expand, which causes velocity fluctuations conceptually similar to the ones of particles in sand. Furthermore, clay aggregates can break apart into smaller pieces, often in the form of individual platelets, thus leading to further meso-related velocity fluctuations. The description of this process is in principle analogous to velocity fluctuations that happen due to particle crushing in granular media and collapse of cellular structures in foams, for which the concept of T_m was introduced in Riley et al. (2023b). Experimental evidence of the pore collapse and the breakage of aggregates is commonly measured by scanning electron microscope images (Hicher et al., 2000; Cotecchia et al., 2019; Guglielmi et al., 2022) and mercury intrusion porosimetry (Delage and Lefebvre, 1984; Tarantino and De Col, 2008; Delage, 2010). Finally, in relating the meso-related temperature T_m to the average kinetic energy of the velocity fluctuations of mesoscopic structures, we notice that T_m must vanish at thermodynamic equilibrium as those motions vanish.

3. Hydrodynamic procedure

Apart from the meso-related entropy s_m , as the thermodynamic conjugate to the meso-related temperature T_m discussed above, we follow Einav and Liu (2018) and consider density ρ , elastic strain ϵ_{ij}^e , and thermal entropy s as additional classical state variables for any porous medium in solid mechanics. Given the so-defined set of state variables $\{\rho, \epsilon_{ij}^e, s, s_m\}$, we follow the rigorous physical principles of the hydrodynamic procedure by Landau and Lifshitz (1980, 1987), and begin by defining the internal energy density u as follows:

$$u \equiv u(\rho, \epsilon_{ij}^e, s, s_m). \quad (1)$$

Provided with this general structure of the internal energy, each of the state variables has a conjugate variable given by:

$$\mu \equiv \frac{\partial u}{\partial \rho}, \quad \sigma_{ij}^e \equiv \frac{\partial u}{\partial \epsilon_{ij}^e}, \quad T \equiv \frac{\partial u}{\partial s}, \quad T_m \equiv \frac{\partial u}{\partial s_m}, \quad (2)$$

with μ being the chemical potential, σ_{ij}^e the elastic stress, and T and T_m the thermal and meso-related temperatures, respectively. The complete derivation of the hydrodynamic procedure can be found in Alaei et al. (2021) for materials that involve exactly the same set of state variables. Following this paper, it is also possible to derive the thermodynamic pressure p^T from the internal energy potential, and relate it to the other state variables:

$$p^T \equiv - \left. \frac{\partial (u/\rho)}{\partial (1/\rho)} \right|_{\frac{s}{\rho}, \frac{s_m}{\rho}, \epsilon_{ij}^e} = \mu \rho + sT + s_m T_m - u. \quad (3)$$

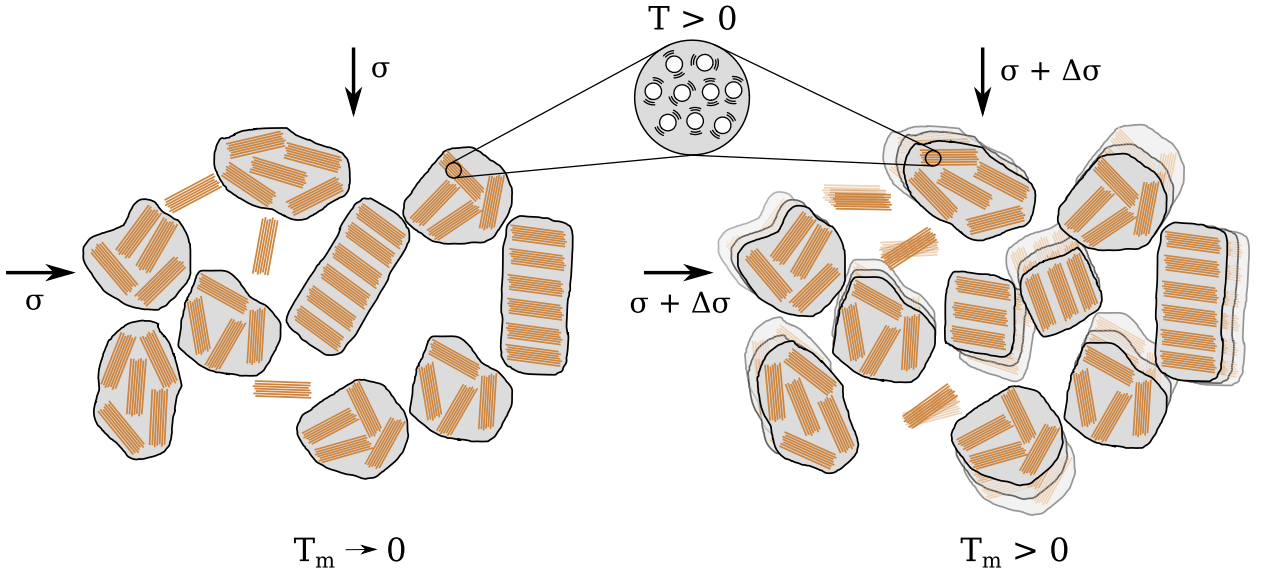


Fig. 3. The two distinct temperatures employed in the hydrodynamic model for clay. Thermal temperature T describes the average fluctuating kinetic energy at the atomic scale, whereas the meso-related temperature T_m represents the average fluctuating kinetic energy of clay aggregates and individual particles.

Importantly, following the hydrodynamic procedure the total stress is always given by:

$$\sigma_{ij} \equiv \sigma_{ij}^e + \sigma_{ij}^d + p^T \delta_{ij}, \quad (4)$$

where δ_{ij} is the Kronecker delta tensor, and σ_{ij}^d is the dissipative viscous stress. This relation has been demonstrated to be highly insightful for the field of geomechanics. In particular, it was proven critical for rigorously retrieving the effective stress of both saturated and unsaturated soils (Jiang et al., 2017; Einav and Liu, 2023), as well as being instrumental in explaining the recurrence of stress serrations in porous media (Riley et al., 2023b). More details on this relation between the above general hydrodynamic stress decomposition and the effective stress principle from soil mechanics are provided in Appendix A, in light of the specific forms of σ_{ij}^d and p^T in Terracotta. Here, the significance of this relation will be further demonstrated by recovering the comprehensively diverse set of rate effects which are manifested in clay.

Finally, notice that during the development of the model itself we will explicitly derive the thermodynamic pressure rather than the chemical potential, as it is this pressure which directly affects the constitutive response through the stress decomposition in Eq. (4). Nevertheless, it is always possible to recover the chemical potential given Eq. (2).

3.1. Evolution of internal variables

Each of the previously defined state variables $\{\rho, \epsilon_{ij}^e, s, s_m\}$ requires its own evolution equation. The evolution of density ρ can be calculated directly from the conservation of mass:

$$\dot{\rho} + \nabla_i (\rho v_i) = 0. \quad (5)$$

where v_i is the velocity vector. In elementary experimental samples in geomechanics, it is prudent to ignore the advection of the density ($v_i \nabla_i \rho \approx 0$). By further identifying the trace of the velocity gradient with the negative volumetric strain rate ($\dot{\epsilon}_v = -\nabla_i v_i$), the rate of density can be found from $\dot{\rho} = \rho \dot{\epsilon}_v$. It is then useful to consider the solid fraction $\phi = \rho / \rho_s$, with ρ_s being the solid density defined as the ratio of solid mass over solid volume. In alignment with existing clay models and the prevalent assumption in experimental geomechanics, we consider solid clay particles to be incompressible with $\dot{\rho}_s \approx 0$. Accordingly, the rate of the solid fraction can be calculated from:

$$\dot{\phi} = \phi \dot{\epsilon}_v, \quad (6)$$

which justifies the corresponding relation in Fig. 1.

Furthermore, the thermal and meso-related entropies satisfy their own balance equation, respectively (Jiang and Liu, 2009):

$$\dot{s} + \nabla_i (s v_i - f_i) = \frac{R}{T} \geq 0, \quad (7)$$

$$\dot{s}_m + \nabla_i (s_m v_i - f_i^m) = \frac{R_m}{T_m}, \quad (8)$$

where sv_i and s_mv_i are the thermal and meso-related convective entropy currents; similarly, f_i and f_i^m are the thermal and meso-related dissipative entropy currents; and finally R and R_m are the thermal and meso-related entropy productions, with only the former required to be positive by virtue of the second law of thermodynamics.

Finally, the elastic strain has its own balance law (Einav and Liu, 2018):

$$\dot{\epsilon}_{ij}^e + v_k \nabla_k \epsilon_{ij}^e + \Omega_{ik} \epsilon_{kj}^e - \epsilon_{ik}^e \Omega_{kj} = \dot{\epsilon}_{ij} - \dot{\epsilon}_{ij}^p, \quad (9)$$

where $\dot{\epsilon}_{ij} = -\frac{1}{2}(\nabla_i v_j + \nabla_j v_i)$ and $\Omega_{ij} = \frac{1}{2}(\nabla_i v_j - \nabla_j v_i)$ are the symmetric and anti-symmetric parts of the velocity gradient, and $\dot{\epsilon}_{ij}^p$ is the dissipative plastic strain rate. Considering negligible rotations and advection in elementary experimental samples, this balance law simplifies to:

$$\dot{\epsilon}_{ij}^e = \dot{\epsilon}_{ij} - \dot{\epsilon}_{ij}^p, \quad (10)$$

which reveals the strain rate decomposition that is frequently used in soil mechanics.

3.2. Rates of entropy productions

By combining the rates of the state variables in Eqs. (5) and (7) to (9) with momentum and energy conservation laws, as well as symmetry requirements, and considering materials with exactly the same set of state variables, the hydrodynamic procedure yields the following general form of the total rate of entropy production (Alaei et al., 2021; Riley et al., 2023b):

$$R + R_m = f_i \nabla_i T + f_i^m \nabla_i T_m + \sigma_{ij}^d \dot{\epsilon}_{ij} + \dot{\epsilon}_{ij}^p \sigma_{ij}^e \geq 0, \quad (11)$$

which is also strictly non-negative. Since the current paper focuses on the local constitutive response, corresponding to elementary laboratory tests, material homogeneity requires the gradient terms $\nabla_i T$ and $\nabla_i T_m$ to vanish, from which:

$$R + R_m = \sigma_{ij}^d \dot{\epsilon}_{ij} + \dot{\epsilon}_{ij}^p \sigma_{ij}^e \geq 0. \quad (12)$$

While not important for the current paper, the gradients of the temperatures must be considered when solving boundary value problems. In particular, dealing with the contribution of $\nabla_i T$ to entropy production is required for an accurate diffusive heat equation (Chen et al., 2023), while $\nabla_i T_m$ introduces a clearly physical length scale, with which localisation features such as shear and compaction bands could be predicted (Riley, 2023).

Next, we consider the principle of two-stage irreversibility first introduced by Jiang and Liu (2009). Accordingly, energy within the system can either flow into the micro-scale, in which case directly elevating the thermal temperature T , or initially induce fluctuating motions at the meso-scale scale by elevating T_m , but subsequently flow to the micro-scale in a two-stage process. Specifically, we consider the following split of entropy productions into the meso-related and thermal scales, respectively:

$$R_m = \sigma_{ij}^d \dot{\epsilon}_{ij} - \bar{\eta} T_m^2, \quad (13)$$

$$R = \dot{\epsilon}_{ij}^p \sigma_{ij}^e + \bar{\eta} T_m^2 \geq 0, \quad (14)$$

where $\pm \bar{\eta} T_m^2$ signifies the energy sink from the meso- to the micro-scale, while $\bar{\eta} \geq 0$ ensures the direction of the energy flow from the higher scales downwards. This means that energy cannot flow backward in this process, and that the clay particles and aggregates are here considered 'athermal', in the sense that thermal collisions of the microscopic atoms within them do not cause those mesoscopic structures to move. Fig. 4 illustrates the concept of two-stage irreversibility based on the specific allocation proposed above of the viscous dissipation $\sigma_{ij}^d \dot{\epsilon}_{ij}$ and plastic dissipation $\dot{\epsilon}_{ij}^p \sigma_{ij}^e$ to the thermal and meso-related scales, respectively.

Notice that adding the last two equations satisfies the first equality in Eq. (12). Also notice that this allocation into the two scales of entropy productions is simpler than the one originally proposed by Jiang and Liu (2009), who have allocated some of the viscous dissipation into the thermal entropy production.

3.3. New reciprocity conditions

Furthermore, unlike previous hydrodynamic models we also introduce a novel form of coupling dissipative processes, which will later be shown to be at the heart of the simplicity of the newly introduced Terracotta model. Most generally, we consider the following relationships for the dissipative stress and plastic strain rate tensors:

$$\sigma_{ij}^d = X_{ijkl}^d \dot{\epsilon}_{kl}, \quad (15)$$

$$\dot{\epsilon}_{ij}^p = X_{ijkl}^p \sigma_{kl}^e, \quad (16)$$

where X_{ijkl}^d and X_{ijkl}^p are newly defined transport coefficients, which may depend on the state variables. These coefficients linearly drive the fluxes σ_{ij}^d and $\dot{\epsilon}_{ij}^p$ by their conjugate forces $\dot{\epsilon}_{ij}$ and σ_{ij}^e , respectively. Those transport coefficients will later be used to allow coupling volumetric and shear dissipative processes.

Given $\bar{\eta} \geq 0$, the non-negativity of the thermal rate of entropy production $R \geq 0$ in Eq. (14) is satisfied by ensuring non-negative plastic dissipation, while the non-negativity of the total rate of entropy production $R + R_m \geq 0$ in Eq. (12) is met by further ensuring non-negative viscous dissipation. Accordingly, we require:

$$X_{ijkl}^d \dot{\epsilon}_{ij} \dot{\epsilon}_{kl} \geq 0, \quad X_{ijkl}^p \sigma_{ij}^e \sigma_{kl}^e \geq 0. \quad (17)$$

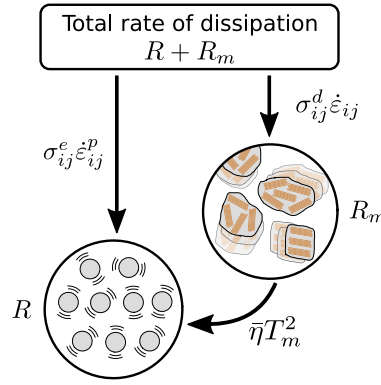


Fig. 4. Concept of two-stage irreversibility (Jiang and Liu, 2009, 2014). The plastic contribution of the total rate of dissipation decays directly to the micro-scale, whereas the viscous contribution decays to the meso-scale to produce meso-related entropy and then further decays to the micro-scale to produce thermal entropy. Notice the subtle difference from the original proposition by Jiang and Liu (2009), where the viscous contribution to the mechanical dissipation was split between the two scales.

Following Onsager (1931), the plastic X_{ijkl}^p and viscous transport coefficients X_{ijkl}^d must satisfy a number of reciprocity requirements, stemming from time-reversal symmetry and positiveness of the dissipation. Specifically, the tensors of transport coefficients must be symmetric with $X_{ijkl}^d = X_{klij}^d$ and $X_{ijkl}^p = X_{klij}^p$. Having symmetric stress and strain rate tensors, the transport coefficients are further required to satisfy symmetries with $X_{ijkl}^d = X_{jikl}^d = X_{ijlk}^d$ and $X_{ijkl}^p = X_{jikl}^p = X_{ijlk}^p$ (De Groot and Mazur, 1954). Moreover, for any given driving force they must be positive semi-definite, in order to ensure the non-negativity of viscous and plastic dissipation expressions, stemming from Eq. (17). By representing the symmetric tensors ϵ_{ij}^p and σ_{ij}^e as 6-tuple vectors, we can reformulate the transport coefficients as 6×6 matrices. To ensure positive semi-definiteness, these matrices must possess only non-negative eigenvalues.

In the model development below we carefully ensure the above requirements by considering the invariants of the various terms. In particular, the viscous transport coefficients X_{ijkl}^d will be simplified to comprise only uncoupled volumetric and deviatoric coefficients, in other words, avoiding any cross-coupling between the volumetric and deviatoric terms. In contrast, the plastic transport coefficients X_{ijkl}^p will include cross-coupling terms between the volumetric and deviatoric coefficients. In fact, this is one of the key technical novelties in this paper compared to previous hydrodynamic models for soils (Jiang and Liu, 2009, 2014; Zhang and Cheng, 2015, 2016), which did not allow for such cross-coupling. In practice, this form of coupling stands behind the success of arriving at such a simple model as the Terracotta one summarised in Fig. 1.

4. The model

The previous two sections have motivated the use of the meso-related temperature as an internal variable in clay, and the general hydrodynamic considerations with which constitutive models could be developed that include this variable. The purpose of the current section is to adopt these considerations to gradually unravel the complete set of the simple equations behind the Terracotta model depicted in Fig. 1.

4.1. Triaxial notation

As implied from Fig. 1, the current form of the Terracotta model is here derived in terms of standard triaxial stress and strain invariants. For example, these invariants include the volumetric stress (pressure) $p = \frac{1}{3} \sigma_{ii}$, the deviatoric stress $q = (\frac{3}{2} \sigma'_{ij} \sigma'_{ij})^{1/2}$, the volumetric strain rate $\dot{\epsilon}_v = \dot{\epsilon}_{ii}$, and the deviatoric strain rate $\dot{\epsilon}_s = (\frac{2}{3} \dot{\epsilon}'_{ij} \dot{\epsilon}'_{ij})^{1/2}$, with σ'_{ij} and $\dot{\epsilon}'_{ij}$ being the deviatoric components of the stress and strain rate tensors, respectively. Finally, to account for the direction of deviatoric components when comparing with triaxial experiments, we consider $q = (\sigma_a - \sigma_r)$ and $\dot{\epsilon}_s = \frac{2}{3}(\dot{\epsilon}_a - \dot{\epsilon}_r)$, where subscripts 'a' and 'r' denotes axial and radial directions in the triaxial conditions.

All other notations follow similar lines, and include the volumetric and deviatoric elastic stresses p^e and q^e , the volumetric and deviatoric viscous stresses p^d and q^d , the volumetric and deviatoric elastic strains ϵ_v^e and ϵ_s^e , and the volumetric and deviatoric plastic strain rates $\dot{\epsilon}_v^p$ and $\dot{\epsilon}_s^p$, respectively. Based on these notations, and invariant structure, all the previous constitutive equations could be specialised for triaxial testing conditions. For example, the stress decomposition in Eq. (4) can be specialised using:

$$p = p^e + p^d + p^T, \quad q = q^e + q^d, \quad (18)$$

while the decomposition of the strain rates in Eq. (10) could be specialised using:

$$\dot{\epsilon}_v^e = \dot{\epsilon}_v - \dot{\epsilon}_v^p, \quad \dot{\epsilon}_s^e = \dot{\epsilon}_s - \dot{\epsilon}_s^p. \quad (19)$$

Similar reformulations of Eqs. (13)–(17) from general tensorial to triaxial convention are considered in the following without explicit exposition.

4.2. Internal energy density

In the proposed model the internal energy density depends on the density, elastic strain and the meso-related entropy. Furthermore, this energy is considered to be additive, and includes the elastic energy u^e associated with the straining of the skeleton and meso-related internal energy u^m associated with the non-equilibrium kinetic energy of the velocity fluctuations:

$$u(\rho, \epsilon_v^e, \epsilon_s^e, s_m) = u^e(\rho, \epsilon_v^e, \epsilon_s^e) + u^m(s_m), \quad (20)$$

where any reliance on the thermal entropy s has been omitted from the above, as we focus on isothermal systems that operate close to room temperature.

Following Riley et al. (2023a) as inspired by Jiang and Liu (2003), Nguyen and Einav (2009), Rubin and Einav (2011) and Alaei et al. (2021), the elastic energy density is made dependent on both the elastic strain and density, and is given by:

$$u^e = \left(\frac{\rho}{\rho_s^*}\right)^6 \left(\frac{\tilde{K}}{6} \epsilon_v^{e^3} + \frac{3}{2} \tilde{G} \epsilon_v^e \epsilon_s^{e^2}\right), \quad (21)$$

with ρ_s^* being the unstressed solid density, and $\tilde{K}$ and $\tilde{G}$ the intrinsic bulk and shear moduli, respectively. This formulation was shown by Riley et al. (2023a) to recover many elastic features of soils, including in broadly predicting elastic wave speeds and instantaneous stiffness. In particular, the specific choice of the power coefficients in the elastic energy density above is taken from the broader family explored by Riley et al. (2023a). In the following section, this choice is justified for clay through predictions against experimental stiffness measurements from the literature.

Apart from the elastic energy density u^e , the definition of the total energy density u in Eq. (20) further requires us to specify the meso-related energy density, which is here taken directly from (Riley et al., 2023b):

$$u^m = \Gamma \frac{s_m^2}{4}, \quad (22)$$

where Γ is a coefficient that was introduced to set the unit of T_m to Kelvin.

4.3. Elastic relationships

Given Eqs. (2), (20) and (21), the mean and deviatoric elastic stress components emerge as partial derivatives of the internal elastic energy density u^e with respect to the corresponding elastic strain invariants:

$$p^e = \frac{\partial u^e}{\partial \epsilon_v^e} = \frac{\phi^6}{2} \left(\tilde{K} \epsilon_v^{e^2} + 3 \tilde{G} \epsilon_s^{e^2} \right), \quad (23)$$

$$q^e = \frac{\partial u^e}{\partial \epsilon_s^e} = 3 \tilde{G} \phi^6 \epsilon_v^e \epsilon_s^e, \quad (24)$$

where we have replaced the ratio of the densities in Eq. (21) by the solid fraction $\phi = \rho/\rho_s$ under the assumption of the incompressibility of the solids $\dot{\rho}_s = 0$ with $\rho_s = \rho_s^*$, as aligned with Section 3.1. Notice that the above results provide the background for the elastic stress components depicted in the first two lines of Fig. 1.

The instantaneous (elastic) bulk K and shear G moduli are obtained through further partial derivation of the elastic stresses with respect to the elastic strains:

$$K = \frac{\partial p^e}{\partial \epsilon_v^e} = \phi^6 \tilde{K} \epsilon_v^e, \quad (25)$$

$$G = \frac{\partial q^e}{\partial \epsilon_s^e} = 3 \phi^6 \tilde{G} \epsilon_v^e. \quad (26)$$

For isotropic elastic states, where $q^e = 0$, so that from Eq. (24) also $\epsilon_s^e = 0$, the instantaneous bulk and shear moduli can be expressed by the following proportionality using Eq. (23):

$$K \propto G \propto \phi^3 \sqrt{p^e}. \quad (27)$$

In thermodynamic equilibrium, when wave measurements are taken, the elastic pressure becomes approximately the total pressure $p^e \rightarrow p$ (Riley et al., 2023a). Accordingly, we notice the dependence on the square-root of the pressure, which agrees with well-known empirical relations for the instantaneous shear stiffness of clays (Hardin and Black, 1968; Jamiolkowski et al., 1995; Mitchell and Soga, 2005; Carlton and Pestana, 2012). To assess the dependence of the instantaneous stiffness on the solid fraction, we compare the above-established elastic relations against two well-known density functions by Hardin and Black (1968) and Jamiolkowski et al. (1995) in Fig. 5, where the function values are normalised by their maximum density values. The dependence of G on ϕ^3 , which follows from the ϕ^6 -scaling of the elastic energy density in Eq. (21), is in good agreement with these empirical relations.

The power coefficient defining the density-dependence of the elastic energy can be taken differently than 6, thus serving to fit any material at hand, as explored by Riley et al. (2023a). For example, using ϕ^3 in the energy, and thus $\phi^{3/2}$ in the stiffness appears most appropriate for sand (Rubin and Einav, 2011; Alaei et al., 2021). However, for clay, the current scaling seems to work very well, as shown by Fig. 5.

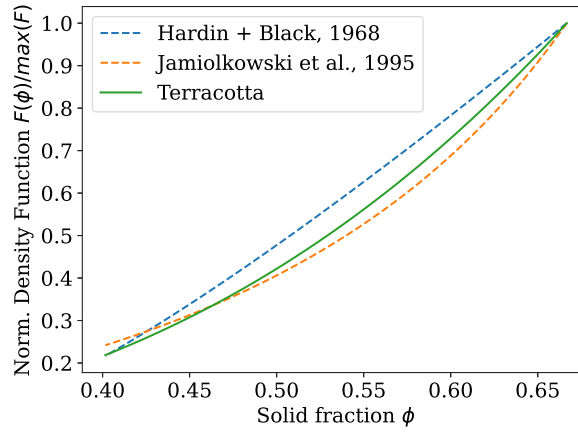


Fig. 5. The functions representing the dependence of instantaneous shear stiffness of two well-known empirical relations (Hardin and Black, 1968; Jamiolkowski et al., 1995) and the derived elastic relationship in Eq. (27). The functions are normalised by their values at the maximum density.

Finally, it is important to note that the elastic energy density in Eq. (21) reveals a physical elastic instability when the determinant of the instantaneous stiffness matrix becomes zero. This instability criterion (Jiang and Liu, 2003; Nguyen and Einav, 2009) can be written in terms of the ratio between the deviatoric and volumetric elastic stresses:

$$\frac{|q^e|}{p^e} \leq M_e, \quad M_e = \sqrt{\frac{3\tilde{G}}{\tilde{K}}}, \quad (28)$$

which in practical terms, puts an upper bound to the attainable elastic stress ratio.

4.4. Dissipative fluxes

The viscous stress and the plastic strain rate represent the dissipative fluxes in the rate of entropy production given by Eq. (12). These are specified by their respective Onsagarian transport coefficients X_{ijkl}^d and X_{ijkl}^p defined in Eqs. (15) and (16), respectively. When formulated in triaxial invariants the general relation for the viscous stress introduced in Eq. (15) becomes:

$$\begin{bmatrix} p^d \\ q^d \end{bmatrix} = T_m \begin{bmatrix} \bar{\alpha} & \bar{\beta} \\ \bar{\beta} & \bar{\gamma} \end{bmatrix} \begin{bmatrix} \dot{\epsilon}_v \\ \dot{\epsilon}_s \end{bmatrix}, \quad (29)$$

where the above coefficients have to satisfy the following Onsager's reciprocity conditions:

$$\bar{\alpha} \geq 0, \quad \bar{\gamma} \geq 0, \quad \bar{\alpha}\bar{\gamma} - \bar{\beta}^2 \geq 0. \quad (30)$$

In pursuit of simplicity, we disregard the possible dependence of the viscous transport coefficients on any state variable other than the meso-related temperature, and set up the cross-coupling term in Eq. (29) to zero in the absence of consistent experimental data for its calibration:

$$\bar{\alpha} = 2\frac{\alpha}{\Gamma}, \quad \bar{\gamma} = 2\frac{\gamma}{\Gamma}, \quad \bar{\beta} = 0, \quad (31)$$

where α and γ represent material parameters, while the dependence of $\bar{\alpha}$ and $\bar{\gamma}$ on $2/\Gamma$ was introduced so that eventually the rate of the meso-related temperature $\dot{T}_m$ will attain the simple form shown in Fig. 1.

Similarly to the above, the plastic strain rate in Eq. (16) can be expressed in terms of the triaxial invariants using:

$$\begin{bmatrix} \dot{\epsilon}_v^p \\ \dot{\epsilon}_s^p \end{bmatrix} = T_m \begin{bmatrix} \bar{a} & \bar{b} \\ \bar{b} & \bar{c} \end{bmatrix} \begin{bmatrix} p^e \\ q^e \end{bmatrix}, \quad (32)$$

where the above coefficients have to satisfy the following Onsager's reciprocity conditions:

$$\bar{a} \geq 0, \quad \bar{c} \geq 0, \quad \bar{a}\bar{c} - \bar{b}^2 \geq 0. \quad (33)$$

It is essential to highlight that in the above we have introduced a linear dependence on the meso-related temperature T^m for both the viscous and plastic fluxes, aligned with the pioneering work of Jiang and Liu (2009, 2014) for granular media. This choice holds a critical significance, ensuring the dissipation fluxes seamlessly vanish as the system approaches equilibrium ($T_m = 0$), irrespective of the strain rates and elastic stresses. It is also worthy to note that any power law of T_m rather than the linear dependence would achieve the same effect. The above choice of just a simple linear dependence is sufficient for the current purpose – not only does it facilitate the model's adherence to equilibrium conditions but also proves instrumental in comprehensively capturing various aspects of rate-dependency.

Finally, we recall that the consideration of the non-zero cross-coupling term $\bar{b}$ is a novel feature of the current model compared with previous hydrodynamic models for granular media (Jiang and Liu, 2009, 2015). It is thanks to this previously ignored term that we are able to derive such a mathematically simple yet realistic constitutive model as Terracotta.

4.5. Meso-related temperature

The meso-related temperature is determined by partial differentiation of the meso-related energy density in Eq. (22) with respect to its energy-conjugate, the meso-related entropy:

$$T_m = \frac{\partial u^m}{\partial s_m} = \frac{\Gamma}{2} s_m. \quad (34)$$

In addition, the energy sink coefficient in the two-stage irreversibility in Eqs. (13) and (14) is considered to depend linearly on T_m :

$$\bar{\eta} = 2 \frac{\eta}{\Gamma} T_m, \quad (35)$$

which could also be written as $\bar{\eta} = \eta s_m$. Notice also that we have introduced a $2/\Gamma$ dependence similar to the viscous transport coefficients in Eq. (31), so that upon combination with Eqs. (8), (13), (29), (31), (34) and (35) we get the third equation in Fig. 1:

$$\dot{T}_m = \alpha \dot{\epsilon}_v^2 + \gamma \dot{\epsilon}_s^2 - \eta T_m^2, \quad (36)$$

where we have used the fact that $\sigma_{ij}^d \dot{\epsilon}_{ij} = p^d \dot{\epsilon}_v + q^d \dot{\epsilon}_s$ in triaxial conditions and considered homogeneous samples by neglecting the convective and dissipative entropy currents.

4.6. Thermodynamic pressure

Considering the additive structure of the internal energy density described in Eq. (20), and the specific expressions of the elastic and meso-related energies in Eqs. (21) and (22), the thermodynamic pressure becomes:

$$p^T = p_e^T + p_m^T, \quad p_e^T = 5u^e, \quad p_m^T = T_m^2/\Gamma, \quad (37)$$

thus including both an elastic term p_e^T resulting from differentiating the elastic energy u^e , and a meso-related term p_m^T arising from differentiating the meso-related energy u^m . It is noteworthy that the thermodynamic pressure may exhibit further contributions when considering additional phases, such as pore fluid and gas, to comprehensively characterise the state of the material under fully or partially saturated conditions (Jiang et al., 2017; Einav and Liu, 2018; Chen et al., 2023; Einav and Liu, 2023).

Notice that the internal elastic energy is approximately proportional to $u^e \propto p^e \epsilon_v^e$ and thus for reasonable elastic strains $\epsilon_v^e \ll 1$, the elastic part of the thermodynamic pressure p_e^T constitutes a small contribution to the total pressure p given by Eq. (18). The same assumption cannot be made for p_m^T , which could reach any value. In pursuit of simplicity, we may thus neglect the elastic contribution of the thermodynamic pressure to the total pressure:

$$p \cong p^e + p^d + p_m^T, \quad (38)$$

which explains the composition of the first line in Fig. 1.

On the other hand, neglecting the contribution of p_e^T to p is not strictly accurate, and thus the complete model which considers this term is inspected in Appendix B. In summary, both models yield almost identical results.

5. Calibration of transport coefficients

In the previous section the general structure of the Terracotta model was established from first principles. Thus far, the development was grounded directly on physics, and did not consider any experimental observation apart from those related to the elastic properties of clay. In particular, details have been summarised on the thermodynamic reciprocity requirements that the plastic transport coefficients must satisfy, but their particular dependence on state variables have not been established. The purpose of this section is to close this gap, so that the Terracotta model could best match conventional experiments in soil mechanics.

5.1. Steady state condition

In order to establish the three independent plastic transport coefficients in Eq. (32), we consider both the stress-density asymptotic behaviour that clay exhibits during *monotonic* isotropic compression as well as its critical state stress-density relation attained during *monotonic* shear testing. These are exactly the same two observations used for setting up constants in critical state soil mechanics (Schofield and Wroth, 1968; Wood, 1990).

Notice that by italicising *monotonic*, we wish to convey that during such loading condition the meso-related temperature in the material is expected to gradually reach a steady state response. The steady state value of T_m could actually be recovered by setting up $\dot{T}_m = 0$ in Eq. (36), from which we find:

$$T_m^s = \sqrt{\frac{1}{\eta} (\alpha \dot{\epsilon}_v^2 + \gamma \dot{\epsilon}_s^2)}. \quad (39)$$

Given this steady state value of T_m , it is possible to determine the plastic transport coefficients $\bar{a}, \bar{b}, \bar{c}$ in Eq. (32) in terms of observations from monotonic isotropic compression and shear tests.

5.2. Isotropic compression tests

The first coefficient $\bar{a}$ is rescued to match observations made during monotonic isotropic compression tests ($\dot{\epsilon}_s = 0$ and $q^e = 0$, from which $\dot{\epsilon}_s^p = 0$). Under such conditions, the elastic deviatoric stress and strain vanish and the clay response converges asymptotically towards a distinct density-pressure attractor, termed the isotropic compression line (ICL) (Terzaghi, 1925; Terzaghi and Peck, 1948; Skempton and Jones, 1944; Schofield and Wroth, 1968; Burland, 1990). As the material develops asymptotically towards the ICL, we expect the following conditions:

$$\dot{\epsilon}_v \rightarrow \dot{\epsilon}_v^p, \quad p^e \rightarrow p_c(\phi), \quad (40)$$

with $p_c(\phi)$ characterising the stress-density relationship on the ICL. This condition is a little bit delicate, since in practice we imagine the total pressure p rather than the elastic pressure p^e to evolve along the ICL. However, we find that the relative difference $(p - p^e)/p$ in practice is rather negligible due to the slowness of typical isotropic compression tests of clay, and thus the second requirement above may be stated as $p \rightarrow p_c(\phi)$.

Using the plastic reciprocity relationships in Eq. (32) under isotropic compression with $q^e = 0$, we find:

$$\dot{\epsilon}_v^p = T_m^s \bar{a} p^e, \quad T_m^s = \sqrt{\frac{\alpha}{\eta}} |\dot{\epsilon}_v|, \quad (41)$$

where the steady state value of T_m^s is taken from Eq. (39) under $\dot{\epsilon}_s = 0$. Since in isotropic compression we also have $\dot{\epsilon}_v > 0$, and given both requirements in Eq. (40), we find:

$$\bar{a} = \sqrt{\frac{\eta}{\alpha}} \frac{1}{p_c(\phi)}. \quad (42)$$

Next, we adopt the bi-logarithmic relation for $p_c(\phi)$ by Hashiguchi and Ueno (1977) to articulate the characteristic density-pressure relation (ICL) in isotropic compression. Originally formulated to relate specific volume $v = V/V_s$ with p_c , the relation is expressed in terms of the solid fraction $\phi = 1/v$, leading to:

$$p_c(\phi) = p_1 \cdot \phi^\lambda, \quad p_1 = (\phi_1^{-\lambda}) \text{ kPa}, \quad (43)$$

where λ represents the slope of the ICL on a bi-logarithmic plot of pressure against solid fraction and p_1 is calibrated with ϕ_1 representing the solid fraction at a reference pressure of 1 kPa. While this relation is consistent with many constitutive models for clay and closely aligns with experimental observations (Butterfield, 1979), particularly within the range of geotechnically relevant pressures, it is important to state some limitations. Notably, upon compression the solid fraction in the model may exceed the physical maximum of unity under extremely high pressures. However, this scenario is unlikely to develop during typical geotechnical pressure ranges, as later substantiated by the calibrated parameters presented in Table 1. Furthermore, this choice does not impact the structure of the model and can easily be replaced if desired.

The resulting coefficient $\bar{a}$ in Eq. (42) along with the definition of the ICL in Eq. (43) and the plastic reciprocity relations in Eq. (32) provide the foundation for the transport coefficient a depicted in Fig. 1. Notice that in that figure we have extracted $p_c(\phi)$ from the individual transport coefficients and placed it before the plastic transport coefficient matrix.

5.3. Critical state shear tests

The two remaining plastic transport coefficients $\bar{b}$ and $\bar{c}$ are determined from monotonic shear testing observations. Unlike isotropic compression, during these tests the shear strain rate is constant and non-zero ($\dot{\epsilon}_s \neq 0$). Following standard observations in soil mechanics, it is well known that, as shearing proceeds monotonically under such conditions, the volume and stress of clay samples asymptotically cease to change:

$$\dot{\epsilon}_v \rightarrow 0, \quad \{\dot{q}^e, \dot{p}^e\} \rightarrow 0. \quad (44)$$

Considering the above, as well as Eqs. (6), (19), (23) and (24), we further observe that

$$\dot{\epsilon}_v^p \rightarrow 0, \quad \dot{\epsilon}_s^p \rightarrow \dot{\epsilon}_s. \quad (45)$$

In order to determine the two remaining plastic transport coefficients $\bar{b}$ and $\bar{c}$, we draw attention to two additional observations that asymptotically develop during monotonic shear testing.

5.3.1. Critical state stress ratio

The second plastic transport coefficient $\bar{b}$ is determined by drawing attention to the observation of an asymptotic stress criterion which emerges during monotonic shear testing. This stress criterion is usually adopted in critical state soil mechanics (Schofield and Wroth, 1968) to relate the deviatoric to the mean stress at critical state:

$$|q^e| \rightarrow M p^e, \quad (46)$$

which is here defined in terms of the elastic stresses, thus representing a generalisation of the effective stress principle in soil mechanics. It has been recently proven, using hydrodynamic principles, that in the absence of viscous stresses and meso-related

temperature, the elastic stress is nothing but the effective stress when considering the additional presence of water and/or air (Jiang et al., 2017; Einav and Liu, 2018).

Considering Eqs. (44) to (46), it is now possible to determine the value of the $\bar{b}$ coefficient at critical state (cs) from Eq. (32):

$$\bar{b}_{cs} = -\frac{\bar{a} \operatorname{sgn}(q^e)}{M}, \quad (47)$$

where $\operatorname{sgn}()$ is the signum function.

Notice that the above value of $\bar{b}$ has been derived under the assumption of critical state conditions. However, during isotropic compression under zero deviatoric elastic stress, the value of $\bar{b}$ must vanish to prevent plastic deviatoric deformations per Eq. (32). Therefore, the final expression of $\bar{b}$ is defined as:

$$\bar{b} = \left(\frac{|q_e|}{M p_e} \right) \bar{b}_{cs} = -\bar{a} \frac{q^e}{M^2 p^e}, \quad (48)$$

which indeed recovers $\bar{b} = \bar{b}_{cs}$ when $|q_e| = M p_e$ from Eq. (46) at critical state. The above relation provides the basis for the off-diagonal plastic transport coefficient depicted on the right hand-side of Fig. 1, where we have factored out $p_c(\phi)$.

5.3.2. Critical stress-density relation

Finally, the third and last plastic transport coefficient $\bar{c}$ is determined to recover the well-known finding of a critical state (cs) stress-density relation which emerges asymptotically during monotonic shear testing:

$$p^e \rightarrow p_{cs}(\phi), \quad p_{cs} = \omega p_c(\phi), \quad (49)$$

where ω is a newly defined constant. Notice that together Eqs. (46) and (49) establish the so-called critical state line (CSL) that is widely adopted in soil mechanics (Schofield and Wroth, 1968; Wood, 1990). Accordingly, projection of the CSL is assumed parallel to the ICL on a double logarithmic plot of solid fraction against pressure. Further notice that the case of $\omega = \frac{1}{2}$ provides a similar prediction to the one provided by the modified cam clay model (Roscoe and Burland, 1968). Finally, notice that the definition above in terms of the elastic pressure represents a slight generalisation of the effective stress principle in soil mechanics, as discussed earlier.

Based on Eq. (32), under critical state conditions the deviatoric plastic strain rate becomes:

$$\dot{\epsilon}_s^p = T_m^s (\bar{b} p^e + \bar{c} q^e), \quad T_m^s = \sqrt{\frac{\beta}{\eta}} |\dot{\epsilon}_s|, \quad (50)$$

where the steady state value of $T_m \rightarrow T_m^s$ has been obtained from Eq. (39) for critical state conditions, whereby $\dot{\epsilon}_v = 0$.

Combining the above relation for the deviatoric plastic strain rate with the critical state stress ratio in Eq. (46), the relation for $\bar{b}$ in Eq. (48), as well as the stress-density relation at critical state in Eq. (49), and since $\operatorname{sgn}(\dot{\epsilon}_s) = \operatorname{sgn}(q^e)$ at critical state, we find:

$$\bar{c} = \sqrt{\frac{\eta}{\gamma}} \frac{1}{M \omega p_c(\phi)} + \frac{\bar{a}}{M^2}. \quad (51)$$

The above result provides the background for the coefficient c and the plastic strain rate given at the bottom of Fig. 1, where $p_c(\phi)$ was factored out and positioned in front of the plastic transport coefficient matrix.

5.4. Reciprocity conditions

As elaborated in Section 3.3, the verification of the thermodynamic Onsager's reciprocal relations is an essential aspect of hydrodynamics. Due the specific choice of the viscous transport coefficients in Eq. (31) where $\alpha > 0$ and $\gamma > 0$, the viscous transport coefficient matrix always satisfies Onsager's reciprocal relations in Eq. (30).

On the other hand, it remains to be confirmed that the equivalent conditions in Eq. (33) on the plastic transport coefficient matrix in Eq. (32) adhere to these relations. Given the specific forms of plastic transport coefficients in Eqs. (42), (48) and (51), we find:

$$\begin{bmatrix} \dot{\epsilon}_v^p \\ \dot{\epsilon}_s^p \end{bmatrix} = T_m \sqrt{\frac{\eta}{\alpha}} \frac{1}{p_c(\phi) M^2} \begin{bmatrix} M^2 & -\frac{q^e}{p^e} \\ -\frac{q^e}{p^e} & \sqrt{\frac{\alpha}{\gamma}} \frac{M}{\omega} + 1 \end{bmatrix} \begin{bmatrix} p^e \\ q^e \end{bmatrix}. \quad (52)$$

With all the constants residing within the diagonal coefficients being positive, the first two conditions in Eq. (33) are automatically met. The third condition in this equation requires the determinant of the matrix to be non-negative, from which we find an upper bound value M_o to the stress ratio q^e/p^e :

$$\frac{q^e}{p^e} \leq M_o, \quad M_o = M \sqrt{\sqrt{\frac{\alpha}{\gamma}} \frac{M}{\omega} + 1}. \quad (53)$$

By ensuring the above, the thermal entropy production would never become negative, as required by the second law of thermodynamics articulated in Eq. (14).

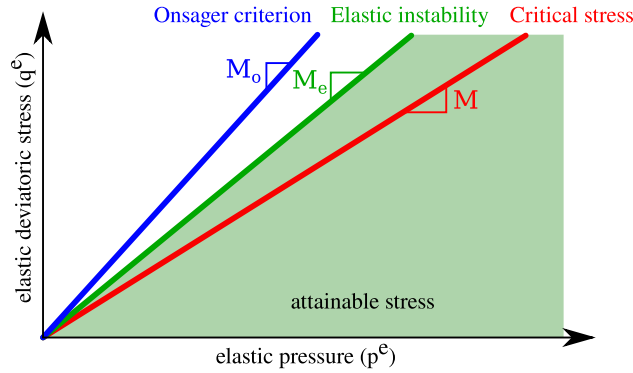


Fig. 6. Important stress ratios in the model, and their relative order.

5.5. The various stress limits

In summary, we recall three important stress ratios. During monotonic shearing, the elastic/effective stress ratio develops asymptotically towards the critical state constant M , as found in Eq. (46). On the other hand, based on the determinant of the elastic stiffness matrix, that stress ratio is also limited by the elastic instability criterion in Eq. (28). Finally, as we have just shown, thermodynamic reciprocity of the plastic transport coefficients as given by the Onsager's determinant relation in Eq. (53) requires the stress ratio to always remain below M_o .

These requirements are ensured by carefully selecting the constants that define them. Notice that according to Eq. (53), M_o is automatically greater than M for the positive coefficients within. Also note that M_e in Eq. (28) limits the elastic stress ratio that can be possibly attained by the model. Therefore, under all circumstances the compliance with the condition in Eq. (53) is guaranteed by requiring the stress ratio from Onsager's determinant relation M_o to always be greater or equal to the elastic instability stress ratio M_e , itself required to always be greater or equal to the critical state ratio M :

$$M_o \geq M_e \geq M. \quad (54)$$

The above stress limits from elastic instability, Onsager's reciprocal relations and critical state are visualised in Fig. 6.

6. Model performance

The preceding sections detailed the rigorous development of the simple Terracotta model for clay, as summarised in Fig. 1. Key to the development was the adoption of the rigorous hydrodynamic procedure and clear physics-based arguments, which gradually revealed the origin of all the equations in Fig. 1. The current section is dedicated to evaluating the model's ability in capturing crucial aspects of the mechanical behaviour of clays. In order to carry out the evaluation the model parameters have to be determined, the equations shown in Fig. 1 must be integrated, and the results obtained should be compared against experimental data from the literature.

In most cases the equations may be explicitly integrated using a sufficiently small time step. This may be adequate for solving the equations under consistent loading rates. However, this may not be adequate for cases with rapidly varying loading rates, as it would require impractically small time steps for a precise solution. To address such cases an implicit integration scheme was favoured and implemented using Python using the implicit multi-step method 'BDF' supplied in `scipy.solve_ivp()` (Virtanen et al., 2020). This approach requires the rate formulation of the state variables, which are derived from the equations given in Fig. 1. Particularly, the rate of the viscous stress involves a 'strain rate acceleration' term $\ddot{\epsilon}$, which we approximate by assuming a constant acceleration during each time step $\ddot{\epsilon} = (\dot{\epsilon}^t - \dot{\epsilon}^{t-1})/\Delta t$.

With respect to the scope of the evaluation, it is useful to notice that the constitutive response of natural clays tends to strongly depend on their diagenesis, as past physio-chemical loading may alter structural properties, including anisotropy and degree of cementation. Addressing these effects would require additional state variables such as fabric tensor in Houlsby et al. (2019) and Amorosi et al. (2021) and/or sensitivity measure in Cotecchia and Chandler (2000). However, including such variables goes beyond the scope of the simple model. Therefore, this section focuses on evaluating the performance of Terracotta using clays which have been reconstituted from their undisturbed natural state. One experiment on undisturbed natural Gytja from Krieg (2000) is further explored as an exception. While this highly compressible and organic soil is not reconstituted, its natural state is understood to be normally consolidated. Furthermore, its internal structure does not show any signs of cementation as evidenced by scanned electron microscope images (Krieg, 2000), justifying the evaluation of Terracotta on this particular natural soil.

Table 1

Terracotta model parameters calibrated for various clays. In the absence of sufficient data to calibrate the rheological constants for the last five clays, unknowns α , γ and η were fixed by the corresponding constants of Kaolin.

	Mechanical constants						Rheological constants		
	$\bar{K}$ [GPa]	$\bar{G}$ [GPa]	M [–]	ω [–]	ϕ_1 [–]	λ [–]	α [K s]	γ [K s]	η [K ⁻¹ s ⁻¹]
Gyttja	20	12	1.30	0.50	0.065	3.62	$5 \cdot 10^5$	$5 \cdot 10^5$	$1.3 \cdot 10^{-2}$
Kaolin	13	5	0.73	0.60	0.267	10.0	$6 \cdot 10^5$	$6 \cdot 10^5$	$1.5 \cdot 10^2$
London clay - B2b	9	6	0.97	0.56	0.309	11.1	$3 \cdot 10^7$	$3 \cdot 10^7$	$7 \cdot 10^{-2}$
Karlsruhe kaolin	11	6	0.95	0.60	0.267	10.0	–	–	–
London clay - A3	26	13	0.90	0.55	0.327	12.5	–	–	–
London clay - C	16	6	0.85	0.62	0.320	11.1	–	–	–
Malaysian kaolin	11	7	0.91	0.65	0.265	11.1	–	–	–
Weald clay	2.6	1.8	0.91	0.50	0.453	17.2	–	–	–

6.1. Calibration of parameters

The purpose of this section is to elucidate the methodology behind the calibration of the Terracotta model parameters (mechanical and rheological), and determine their values for various clays for model evaluation in the following sections. One key benefit in using terracotta is that all of its six mechanical constants are already known to engineers, hence their calibration follows a standard procedure. For example, the calibration of its two elastic constants, that are the bulk and shear moduli $\bar{K}$ and $\bar{G}$, may be done through conventional elastic wave measurements such as the consistent procedure proposed by Riley et al. (2023a), which is applicable to the form of elastic energy by Terracotta in Eq. (21). In scenarios where such measurements are not available, as with the experiments discussed throughout the section, the elastic parameters can alternatively be calibrated by fitting the model to observed stress–strain responses. Moving on to the remaining mechanical constants, we recall that λ and $p_1 = (\phi_1)^{-\lambda}$ kPa define the isotropic compression line (ICL), as described in Section 5.2, and can be directly calibrated using isotropic compression tests. The slope of the ICL (that is, λ) may also be calibrated from oedometric compression tests, as it commonly aligns parallel to the ICL on bi-logarithmic pressure–density plots. The critical stress ratio M and the position of the critical stress–density relation ω can be derived using triaxial compression tests. To enable the model evaluation in the forthcoming sections, the mechanical parameters were calibrated using the outlined standard methodology for various clays, and are listed in Table 1. Notice that p_1 in Fig. 1 can be derived from Eq. (43).

Apart from mechanical constants, three other constants determine the rheological performance of Terracotta. Unlike the six mechanical constants, the three rheological ones are not standard. However, in the following we consider the two viscosity constants to be equal, so the actual number of rheological constants essentially reduces to two: the viscosity constant $\gamma (= \alpha)$ and the meso-related relaxation constant η . Next, in order to understand the phenomenological significance of the two new rheological constants, a simple parametric study is carried out to study the sensitivity of the constitutive response to their alteration on two types of undrained triaxial compression tests, while keeping the mechanical constants fixed for Kaolin in Table 1.

In the first example, the significance of the rheological constants is illustrated in terms of a two-stage monotonic undrained triaxial compression test involving monotonic shear at constant strain rate followed by a long period of stress relaxation. To unpack the effects of the rheological constants on this particular loading scenario, constants γ and η are varied in the following ways. First, to isolate the relaxation stage from the preliminary response, we use a combination of constants $\gamma (= \alpha)$ and η that maintains the same stationary viscous stress at a given strain rate by keeping $\gamma \propto \eta^{1/3}$. As demonstrated in Fig. 7(a,b), it follows that the rate of stress relaxation can be directly tuned by the two rheological constants, while not influencing the first stage of the loading. Second, Fig. 7(c,d) presents the constitutive response for the same type of two-stage tests for varying γ and η while maintaining $\gamma \propto \eta^{-1}$. Although the constants change, we observe that the rate of relaxation remains identical under such fixed proportionality. This is because $\gamma\eta$ controls the rate of change of the meso-related temperature, as would be visualised in the next example. Furthermore, we see that the increase of γ above a certain value visibly increases the total stress during the first monotonic loading phase.

In the second example, we look at the role of the rheological parameters for the case of strain rate jumps, again using undrained triaxial compression. To this end, after reaching a stationary steady state, the strain rate is first increased by tenfold and subsequently decreased back to the initial rate. To systematically identify the effect of changing the rheological parameters, the same two approaches as in the first example are used to vary γ and η . First, by maintaining $\gamma \propto \eta^{1/3}$ Fig. 8(a) shows that the model response to strain rate jumps ranges from a seemingly immediate to a delayed convergence to the new stationary state. Fig. 8(b) demonstrates that higher values of the rheological constants accelerate the convergence of the meso-related temperature T_m to its new stationary state and consequently, thus explaining the faster attainment of the new stationary stress state with increasing rheological constants. Second, Fig. 8(c) reveals that under fixed $\gamma\eta$ the magnitude of the total stress can be controlled by altering the extent of the viscous stress. The shape of the transient response to changes in the strain rate is not affected since the time-variation of the meso-related temperature is kept identical during the loading, as shown in Fig. 8(d).

Having isolated the significance and phenomenological role of the rheological constants, it is now possible to use this understanding to calibrate those constants by best matching the theory to test data. To this end, the stationary value of the stress after strain rate jumps had been used to set the proportionality between $\gamma (= \alpha)$ and η , while transient stress response during relaxation or following strain rate jumps were then used to determine the actual values.

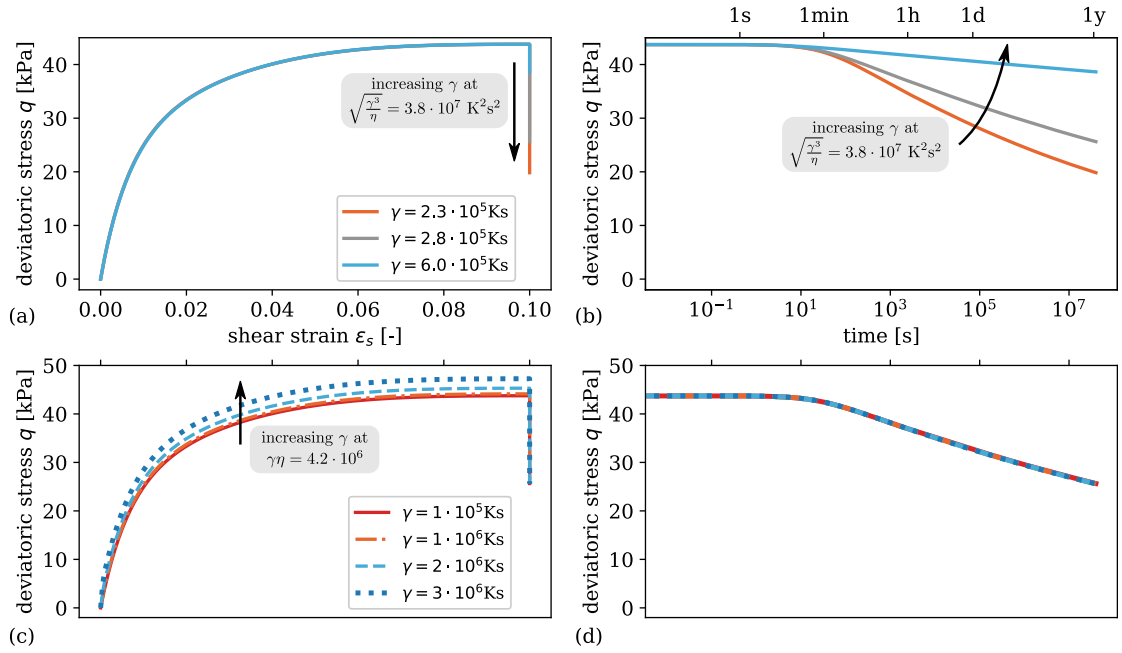


Fig. 7. The impact of the rheological constants on rate hardening and stress relaxation following monotonic undrained triaxial compression. To this end, (a) and (c) show stress-strain response, while (b) and (d) show stress relaxation over time, where η is varied under either constant $\sqrt{\gamma^3/\eta} = 3.8 \cdot 10^7 \text{ K}^2\text{s}^2$ for (a) and (b); or under constant $\gamma\eta = 4.2 \cdot 10^6$ for (c) and (d). The initial state in all simulations is given by $p^e = 100 \text{ kPa}$, $q^e = 0 \text{ kPa}$, $\phi = 0.423$ and $T_m = 0$, while $\dot{\epsilon}_s = 2 \cdot 10^{-5} \frac{1}{s}$ during the deformation stage. The mechanical constants are given by Kaolin in Table 1.

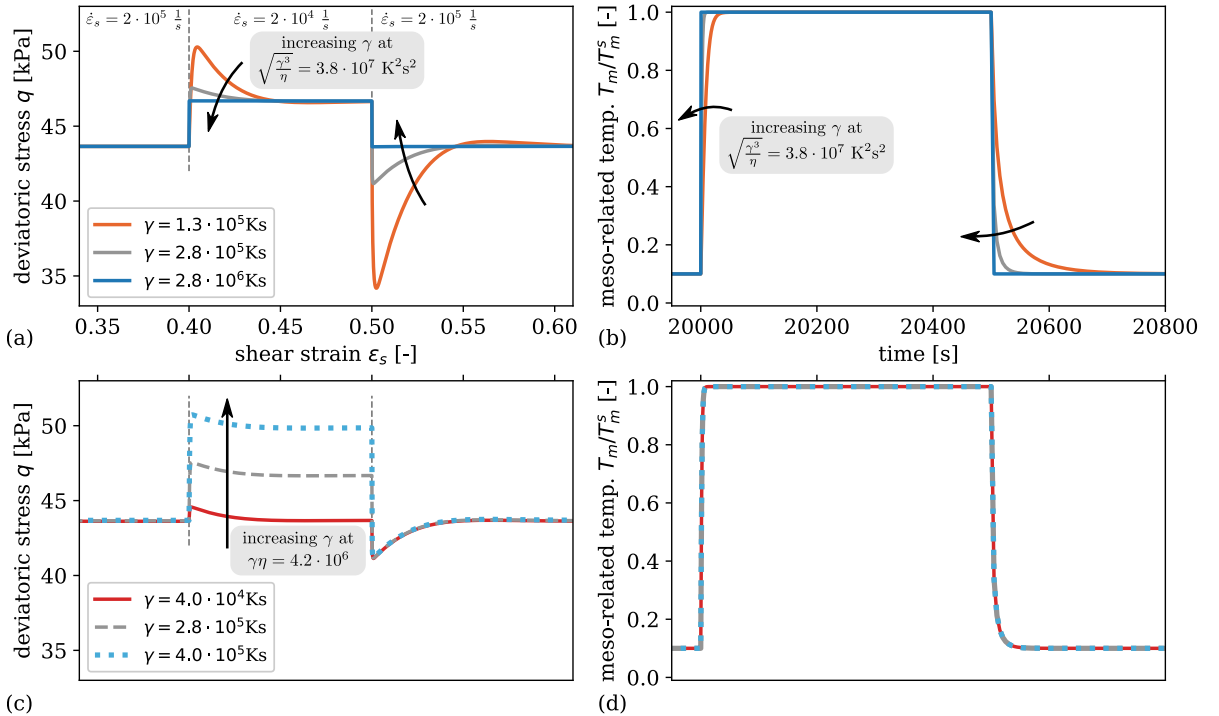


Fig. 8. The impact of the rheological constants on the model response after shear strain rate jumps (as indicated on the figure) during undrained triaxial compression tests. To this end, (a) and (c) show stress-strain response, while (b) and (d) show the meso-related temperature over time, where η is varied under either constant $\sqrt{\gamma^3/\eta} = 3.8 \cdot 10^7 \text{ K}^2\text{s}^2$ for (a) and (b); or under constant $\gamma\eta = 4.2 \cdot 10^6$ for (c) and (d). The remaining (mechanical) constants represent the Kaolin given by Table 1, while the initial state is given by $p^e = 100 \text{ kPa}$, $q^e = 0 \text{ kPa}$, $\phi = 0.423$ and $T_m = 0$.

Table 2

Ensuring the right order of stress limits in Eq. (54) for the various clay parameters in Table 1, as implied from Onsager's reciprocity and elastic instability restrictions.

	$M \leq$	$M_e \leq$	M_o
Gyttja	1.30	1.34	2.47
Kaolin	0.73	1.07	1.09
London clay - B2b	0.97	1.41	1.60
Karlsruhe kaolin	0.95	1.28	1.53
London clay - A3	0.90	1.22	1.46
London clay - C	0.85	1.06	1.31
Malaysian kaolin	0.91	1.38	1.41
Weald clay	0.91	1.44	1.53

Wherever possible, we used available experimental data to calibrate the parameters. Nevertheless, in some cases, the reported data were insufficient, particularly for calibrating the rheological parameters α , γ and η . For simulations specifically related to the experiments in Section 6.3, these parameters were fixed by the rheological constants of kaolin. This choice ensures the model exhibits apparent rate-independence during monotonic loading under the applied strain rates.

Further notice that the London clay subclasses A3, B2b and C represent reconstituted materials from different lithological layers as defined by Gasparre (2005). Although the London clay specimens were reconstituted, it was found that their geological alteration through diagenesis has a substantial effect on the observed behaviour and justifies an independent calibration of the material constants for each given lithology (Gasparre, 2005). Moreover, consider that while the three kaolin soils share the mineral kaolinite, they are indeed distinct materials (as seen through their markedly different Atterberg limits (Wichtmann and Triantafyllidis, 2018; Duque et al., 2022; Tatsuoka et al., 2002)).

Finally, the required order of the stress limits in Eq. (54) is assessed based on Eqs. (28), (46) and (53). As listed in Table 2, under all circumstances the calibrated parameters in Table 1 consistently fulfil the rigorous requirements by Onsager's thermodynamic reciprocity and the one by elastic instability in Eq. (54).

6.2. Apparent rate-independence

Previous hydrodynamic models have incorporated the principle of two-stage irreversibility (Jiang and Liu, 2009) in order to reflect the process of energy flow across material scales. Accordingly, models such as granular solid hydrodynamics for sand (Jiang and Liu, 2009, 2014) unravel a range of rate-dependent phenomena that go beyond the predictions by classical plasticity models. Since the mathematical structure of the Terracotta model builds on similar principles, it is important to start by distinguishing a key feature of rate-dependence for the forthcoming discussion.

To this end, we draw attention to Fig. 9, which just like Fig. 7 presents simulations using Terracotta of undrained triaxial compression tests followed by a year long period of stress relaxation. However, here this is done for a fixed set of rheological constants (for Kaolin) yet under different strain rates, each followed by a one-year relaxation stage. From that figure we see that the stress-strain response of the simulation with the highest strain rate distinctly deviates from the other three simulations under the lower rates. In return, all the lower rate simulations follow a common stress-strain trend during the loading stage. Following the traditional line of thinking, it might be tempting to imagine that the response of these three lower-rate tests is rate-independent, but this would not be entirely accurate. In fact, while all these tests exhibit similar stress-strain curves, their preceding relaxation stage underscores a significant influence of the prior loading rate, resonating with experimental findings as noted by Lacerda and Houston (1973). Therefore, the material embodies within it a sort of rate-dependent memory. In Terracotta, this memory is imprinted through different values of steady-state meso-related temperatures during undrained shearing $T_m^s = \sqrt{\frac{\beta}{\eta}} |\dot{\epsilon}_s|$ (see Eq. (50)), which linearly depend on the deviatoric strain rate. In order to highlight this subtle feature, we will denote the common stress-strain trend of the slow rate tests to be only 'apparently rate-independent'.

The forthcoming analysis will focus on assessing the model's capacity to accurately reflect constitutive responses that are (only) apparently rate-independent.

6.3. Experiments with apparent rate-independence

To study apparently rate-independent constitutive responses, we considered four distinct clays: Weald clay (Henkel, 1956; Parry, 1960), London clay (Gasparre, 2005; Gasparre et al., 2007), Malaysian kaolin (Duque et al., 2022) and Karlsruhe kaolin (Wichtmann and Triantafyllidis, 2018). Here, the only database that included experiments under different constant applied strain rates is the one on Karlsruhe kaolin through undrained triaxial compression. The corresponding differences in the stress-strain behaviour between the different strain rate tests lie well within the experimental scatter shown through repetitive tests under the same conditions. Therefore, that study remains inconclusive as to whether there is a discernible influence of the applied strain rate within the tested range, so in the following the evaluation of Terracotta is carried out by considering apparent rate-independence. Since the experiments on the Malaysian kaolin and the London clay were conducted under substantially slower strain rates ($\dot{\epsilon}_s \approx 3 \cdot 10^{-6}$ and $6 \cdot 10^{-8} \text{ s}^{-1}$, respectively) than those on the Karlsruhe kaolin ($\dot{\epsilon}_s \approx 9 \cdot 10^{-6} \text{ s}^{-1}$), we also consider their response as being apparently

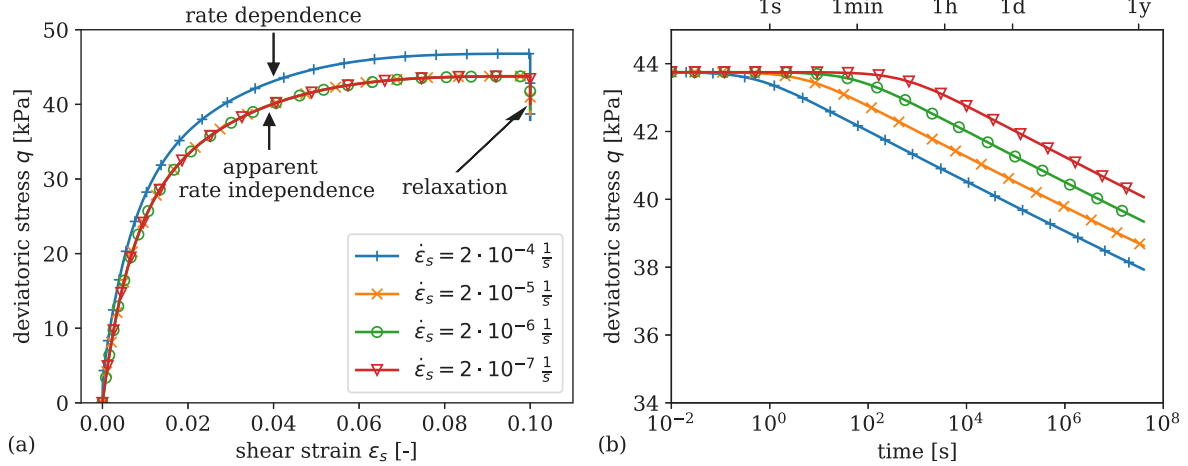


Fig. 9. The concept of ‘apparent rate-independence’, as motivated by two-stage simulation tests involving an undrained triaxial compression stage under different strain rates followed by a relaxation stage over a year. For illustration purposes, the parameters have been calibrated for Kaolin and given by Table 1. The initial state prior to shearing is given by $p^e = 100$ kPa, $q^e = 0$ kPa, $\phi = 0.423$ and $T_m = 0$. (a) stress–strain response. (b) stress relaxation over time.

rate-independent. Note that the applied strain rate was not reported in the experiments on Weald clay and we therefore also assume an apparent rate-independence.

The material parameters for these clays are listed in Table 1. Recall that in the absence of sufficient rate-dependent test data, the rheological constants for these clays were simply fixed by those of kaolin. Irrespective of their exact values, and as demonstrated in Fig. 9, the main point of those adopted values was to ensure that under strain rates lower than $2 \cdot 10^{-5} \text{ s}^{-1}$, which is well above the strain rates applied in the monotonic experiments, the stress–strain response would remain apparently rate-independent. In this way, the actual values of the rheological constants do not have any visible bearing on the constitutive response under constant strain rate.

All simulations commenced from a state of thermodynamic equilibrium ($T_m = 0$). Notice that this assumption ignores the dynamics of the preparation protocol preceding the triaxial shearing conditions. In theory, the attainment of thermodynamic equilibrium requires an infinite period of relaxation, whereby no deformation is imposed on the sample. In practice, there is always a finite time between sample preparation and triaxial loading. However, in the absence of the complete time history of the loading, we assume sufficient time such that the initial state would be in equilibrium. Furthermore, it follows from Eqs. (29) and (37) that the meso-related thermodynamic pressure p_m^T and the viscous stresses p^d and q^d are also zero at thermodynamic equilibrium. Therefore, the elastic stress is given by the experimentally measured total stress, and the elastic strain can be retrieved directly from it, given the solid fraction from the experiment.

Fig. 10 displays the isotropic compression tests carried out on London clay by Gasparre (2005). As shown, the model exhibits a generally good agreement with the experimental data. Notice that the researcher expressed some difficulties in maintaining a constant strain rate during the experiment, particularly in the case of the A3 specimen (depicted in blue), which may reason the deviations observed between the model and the material responses. It is crucial to note, that this isotropic compression test provides an isolated assessment of the plastic transport coefficient $\bar{a}$, as defined in Eq. (42). In short, the model successfully captures the characteristic density–pressure relations evident in the experimental data, thus supporting the method for determining the $\bar{a}$ plastic transport coefficient.

Next, we examine the model’s ability to capture behaviour of clay under different forms of triaxial shearing. To this aim, Fig. 11 shows four experiments on Weald clay, including: (1) a normally consolidated triaxial undrained test, (2) a normally consolidated drained test, (3) an over-consolidated undrained test and (4) an over-consolidated drained test. The initial solid fractions were chosen based on the isotropic compression test reported in Henkel (1956). Notably, the tests did not reach a unique stress–density relation at critical state. Since the critical state of Terracotta under apparent rate-independence is defined uniquely by a stress–density relation, the simulations slightly diverge from the experiments. On the other hand, it is important to highlight accurately measuring the initial density of clays is delicate and error prone. It is therefore possible that the experimental deviation of the final density from a unique stress–density critical state line might have arisen due to this initial bias. It is possible to show that should those tests have indeed reached a unique critical stress–density relationship, as expected by critical state soil mechanics, the comparison would have been even more striking. In conclusions, the model successfully captures the stress–strain response of Weald clay under both undrained and drained conditions and accurately reflects the transition from contractive behaviour of normally consolidated soils to dilative behaviour of over-consolidated soils. These results also reaffirm the method of defining the plastic transport coefficients $\bar{b}$ and $\bar{c}$, respectively defined in Eqs. (48) and (51). In particular, they showcase the model’s ability to develop asymptotically towards the critical state line, and in capturing both the critical stress ratio and the critical density–pressure relation.

To further justify the calibration of the plastic transport coefficients $\bar{b}$ and $\bar{c}$, the evaluation is extended to examine undrained triaxial compression tests on three additional clays, including Malaysian kaolin (Duque et al., 2022), Karlsruhe Kaolin (Wichtmann

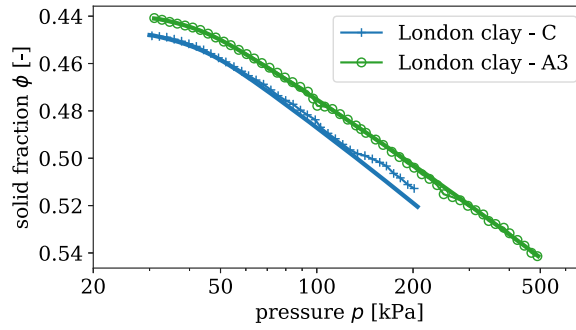


Fig. 10. Isotropic compression tests on London clay. Marked lines represent the experiments by Gasparre (2005) and solid lines the model simulations with parameters given by Table 1.

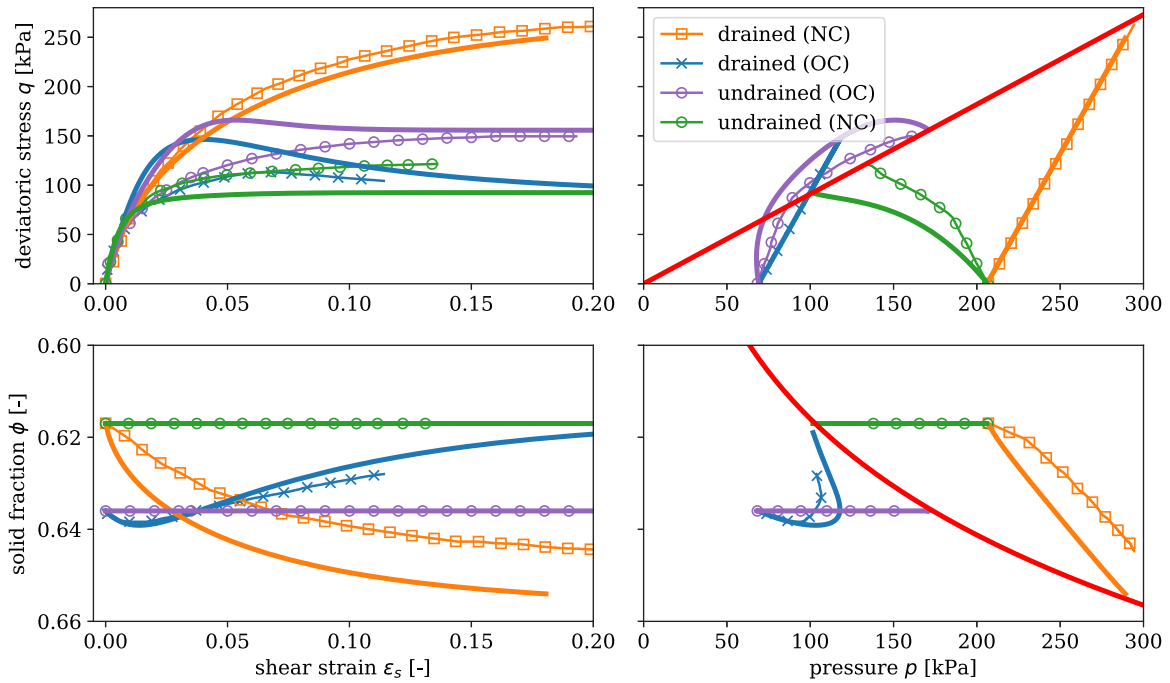


Fig. 11. Constitutive response of Weald clay (Parry, 1960) during drained and undrained triaxial compression tests. The marked lines represent digitised experimental data taken from soilmodels.com (Gudehus et al., 2008), while the solid lines represent the model response based on parameters from Table 1, apart from the solid red lines which denote the critical state relation.

and Triantafyllidis, 2018) and two subclasses of London clay (Gasparre, 2005). The results of the evaluation of both initially normally- and over-consolidated isotropic cases, are presented in Fig. 12. For the London clay and the Malaysian kaolin, simulations began with the same initial densities reported in their respective studies. However, given the considerable variability in initial void ratio measurements for Karlsruhe Kaolin (Wichtmann and Triantafyllidis, 2018), the initial states for normally consolidated samples were relied upon the calibrated ICL, while the initial states for the over-consolidated samples were estimated by projecting their critical states in stress space onto the CSL in the density-pressure plot.

The model demonstrates a strong capability in replicating the responses of both normally and over-consolidated tests across all clays, and consistently with critical state soil mechanics principles all the shear tests converge to the critical state line. In the case of normally consolidated tests, the stress paths by the model align exceptionally well with the empirical data. While the stress paths for the over-consolidated samples generally follow the expected trends, the model does not perfectly match the lines when the mean pressure appears to have an immediate tendency to increase.

It is worth noting that the softening observed in the experimental curves for both the Malaysian and Karlsruhe kaolin clays could be attributed to the onset of strain localisation. In that case, the problem can no longer be idealised through the homogeneous constitutive response of a single material point.

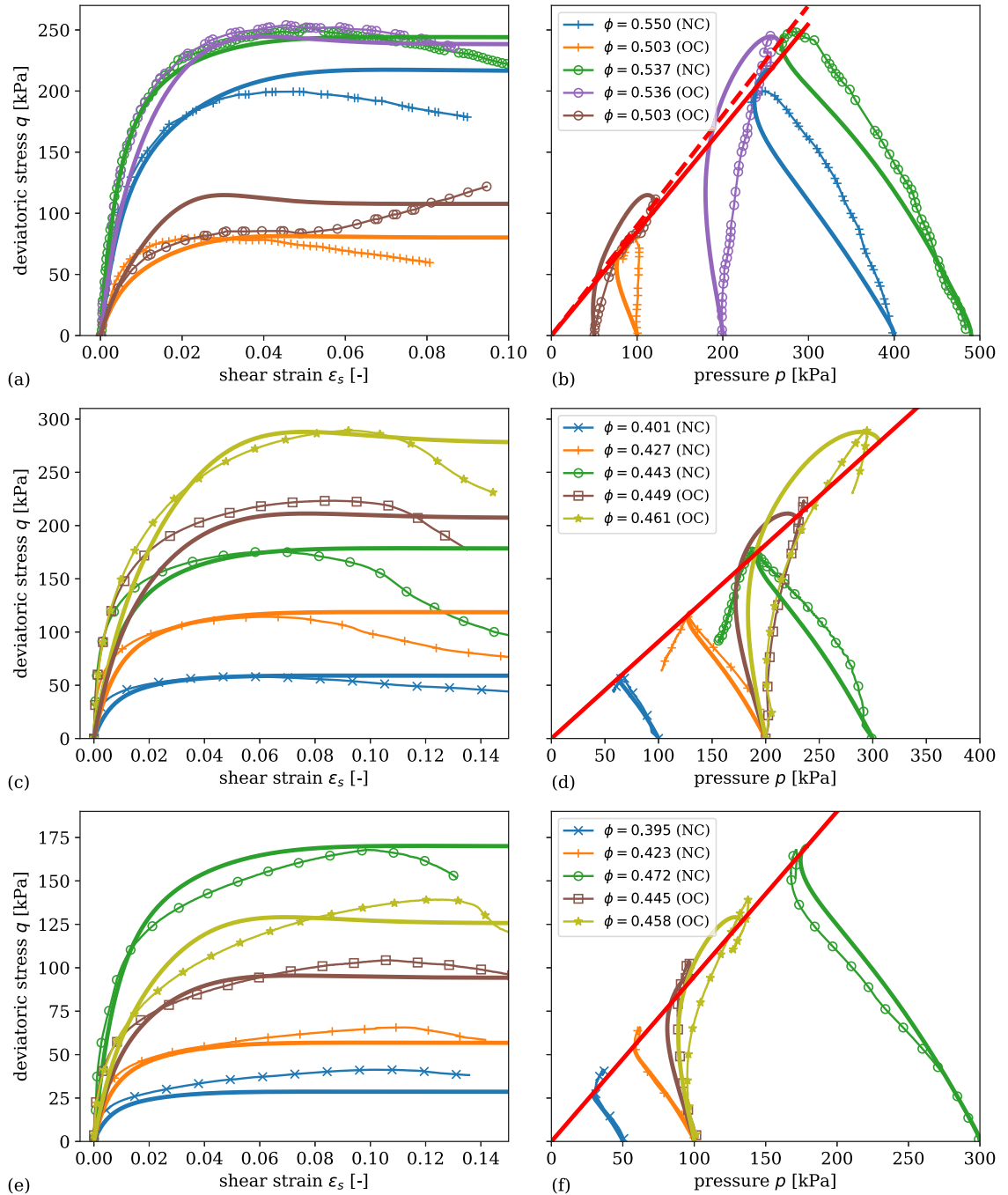


Fig. 12. Undrained triaxial compression tests for different clays, with the constant solid fractions during shear indicated in the legends. The marked lines represent the experiments and the solid thick lines the simulations, apart from red lines which show the corresponding critical state relations. “NC” and “OC” indicate normally- and over-consolidated samples, respectively. (a), (b): London Clay (Gasparre, 2005) where the $\circ$ and $+$ markings denote experimental data for subclasses A3 and C, respectively. (c), (d): Malaysian clay (Duque et al., 2022). (e), (f): Karlsruhe kaolin (Wichtmann and Triantafyllidis, 2018).

6.4. Rate-dependent experiments

In the preceding section, the apparent rate-independence of clays under low-rate monotonic loading conditions was examined. Yet, clays are inherently rate-dependent, often exhibiting pronounced stress–strain phenomena upon sudden variations in the loading rates. This section assesses how the model captures the corresponding phenomena. In that respect, unlike Terracotta, most

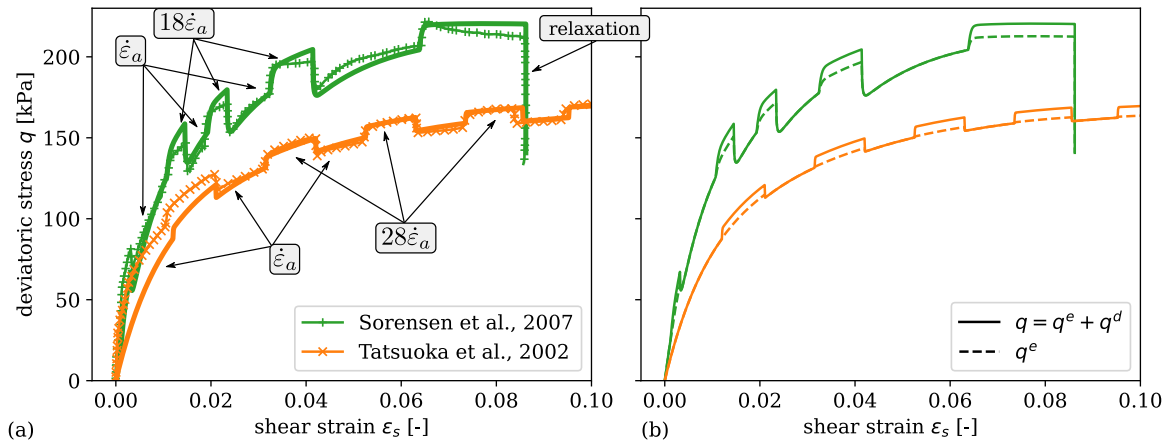


Fig. 13. Undrained triaxial compression tests with multiple step changes of the strain rate from [Tatsuoka et al. \(2002\)](#) and [Sorensen et al. \(2007\)](#). (a) Experiments (marked lines) and simulations (solid lines). The basic axial strain rates $\dot{\epsilon}_a$ are $1.4 \cdot 10^{-7}$ 1/s and $1 \cdot 10^{-5}$ 1/s for the tests from [Sorensen et al. \(2007\)](#) and [Tatsuoka et al. \(2002\)](#), respectively. (b) The corresponding evolution of the total deviatoric stress and its elastic component in the proposed model.

constitutive models designed for rate-dependent clays are tailored to address a specific form of rate-dependency known as ‘isotach’ behaviour ([Šuklje, 1969](#)). Within this past modelling paradigm, isotachs were considered as characteristic loading rate-dependent curves of the stress–strain response. When the loading rate undergoes a change, the material’s response transitions rapidly between these isotachs. In the literature, various characteristic rate-dependent responses exist for various clays and loading rates, as outlined in [Tatsuoka et al. \(2008\)](#). One of these characteristics involves an immediate stress jump following a change in the loading rate, succeeded by a relaxation towards a residual rate-dependent stress–strain line. This phenomena was termed ‘combined viscous’ behaviour in [Tatsuoka et al. \(2008\)](#).

We start by assessing the model’s ability to depict these two distinct characteristic rate-dependent responses of clays. The first involves a reconstituted and normally consolidated kaolin clay exhibiting isotach behaviour during undrained triaxial compression, as documented in [Tatsuoka et al. \(2002\)](#). The second case involves London clay from the B2b subclass, as detailed in [Gasparre et al. \(2007\)](#), which displays minimal rate-dependence during isotropic compression but demonstrates the so-called combined viscous behaviour during undrained triaxial compression, as reported in [Sorensen et al. \(2007\)](#). To this end, the material parameters for the simulations were calibrated using the available data sets in [Tatsuoka et al. \(2002\)](#) and [Sorensen et al. \(2007\)](#), with the chosen values reported on [Table 1](#). Notice that some of those parameters had to be assumed due to the lack of data availability. This, however, is not crucial to the concerned evaluation, which focuses on the qualitative capabilities of the model. In particular, the rheological parameters α , γ and η were calibrated in order to fit the stress–strain response to the strain rate jumps from the experiments.

The left panel of [Fig. 13](#) evaluates the qualitative ability of Terracotta to predict the experimental stress–strain responses of the two different clays in [Tatsuoka et al. \(2002\)](#) and [Sorensen et al. \(2007\)](#) under undrained triaxial compression tests. As shown, the so-called isotach behaviour of the kaolin (in orange) can be captured quite accurately, while the only notable difference is the initial stiffness that is too low in the model prediction. Similarly, the model successfully reproduces the so-called combined viscous behaviour observed in the London clay test (in green). Specifically, the model performs quite well during step changes from high to lower strain rates, depicting stress drops and subsequent relaxation to a higher stress level with ongoing deformation. Given the chosen parameter set, the model does not fully capture the stress relaxation following jumps from slow to faster strain rates. Nevertheless, the model can qualitatively reproduce such effects, as discussed previously in [Section 6.1](#). Finally, notice also that at the end of the shearing stage of the experiment involving the London clay, a four-day relaxation stage followed. Strikingly, the model accurately reflects the resulting extent of the stress drop. Unfortunately, the time evolution of the relaxation was not reported. However, the capability of the model to capture creep and relaxation would be assessed in the next subsection, showing again remarkable qualitative agreement with available experimental data.

The right panel of [Fig. 13](#) examines the conceptual additive structure of the stress in the model. In particular, this panel allows us to illustrate the source of the differences between the characteristic rate-dependency from the viewpoint of the model, by examining the contribution of the elastic deviatoric stress q^e to the total deviatoric stress q . Here, the extent of the viscous deviatoric stress may be assessed by their difference $q^d = q - q^e$. In the case of the so-called isotach behaviour, the model predicts a sudden increase of the viscous stress to accommodate the step change in the strain rate while the elastic stress is not visibly impacted. This is contrasted with the so-called combined viscous behaviour which is accompanied by the model through visible changes in both the elastic and the viscous stresses. The key physics underpinning the differences between these two distinct responses is the evolution of the meso-related temperature T_m , which controls the plastic strain rate. For a given imposed strain rate, the response of the viscous and elastic stresses is thus determined by the rheological parameters α , γ and η , which are responsible for letting T_m reach its new steady state faster or slower.

The isotropic compression test reported on the same London clay by [Sorensen et al. \(2007\)](#) is shown in [Fig. 14](#). This experiment is controlled by the application of a constant stress rate with step changes between 3 and 1 kPa/h. Indeed, this difference in the loading

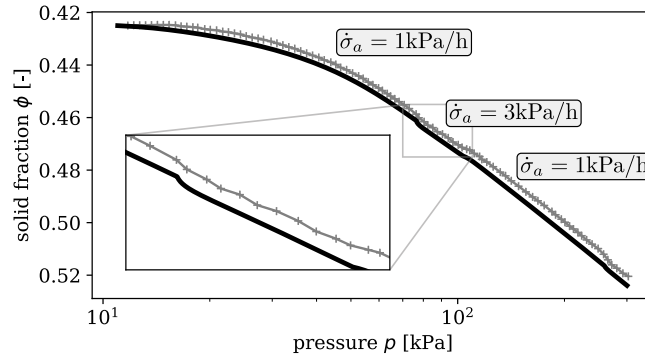


Fig. 14. Comparison of isotropic compression test responses to rapid step changes in the stress rate between experiment (from Sorensen et al., 2007, grey line with symbols) and model simulation (solid black line). The inset highlights the two step changes.

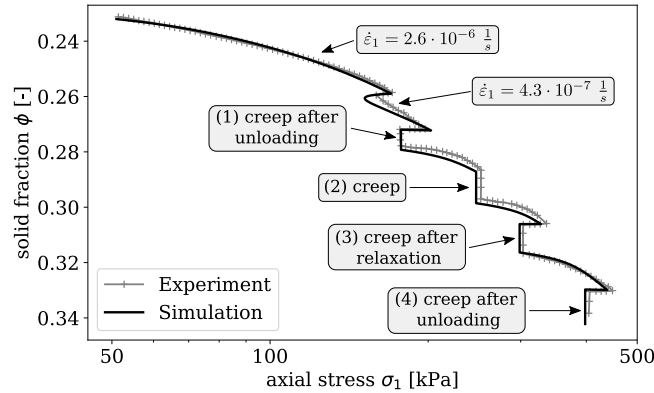


Fig. 15. Comprehensive oedometer test with one change in loading rate and different stages of unloading, relaxation and creep from Niemunis and Krieg (1996) and Krieg (2000).

rate is apparently almost negligible for this particular clay as the volumetric behaviour is only slightly impacted. Nevertheless, the model well captures the first slowing down of the stress rate (as highlighted in the zoomed inset), albeit a slight divergence from the experiment at higher stresses. This divergence may be due to two reasons. The first is that the meso-related temperature never actually reaches its steady state value in the simulation, as the strain rates constantly evolves due to the applied constant rate of stress. The second is that for the current simple model version we have neglected any potential dependencies of the viscous transport coefficients on state variables other than T_m . For example, incorporating pressure dependent rheological coefficients might improve the agreement between the model and this particular experiment. However, this tuning goes beyond the purpose of this current paper, which is to set up as simple as possible hydrodynamic model for clay.

6.5. Relaxation and creep

Aside from material response to changes in loading rate discussed in the previous section, two other well-known forms of the manifestation of rate-dependency in clay include creep and relaxation processes. The latter form has already been illustrated as a preliminary means of highlighting the concept of apparent rate-independence in Section 6.2. In that section we showed that despite the stress–strain response during shear in Fig. 9 exhibiting apparent rate-independence over various strain rates, the stress relaxes differently once the shearing stops, whereby specimens previously subjected to slower shearing rates exhibit correspondingly slower stress relaxation. This difference has been explained through the difference in the steady-state value of T_m with the applied strain rate during the past shear as given by Eq. (39), where the faster-sheared experiments commence with higher T_m 's, thereby requiring longer time to relax or creep the samples to similar levels of stress or strain, respectively.

To assess the model's ability to simulate creep and relaxation, we draw attention to the comprehensive oedometric compression test on a highly organic Gytja reported in Niemunis and Krieg (1996). While further including phases of strain rate jumps, unloading and relaxation, this test primarily focused on exploring the creep behaviour over time as illustrated in Figs. 15 and 16. In particular, this exhaustive (and useful) test initiated with a transition from a higher to a lower loading rate, followed by small unloading, creep, loading, again creep, loading, relaxation, creep, further loading, unloading and creep, as delineated in Fig. 15. The parameters used for the simulation are detailed in Table 1. As described by Krieg (2000) all creep stages lasted 10 days. Fig. 15 showcases Terracottas

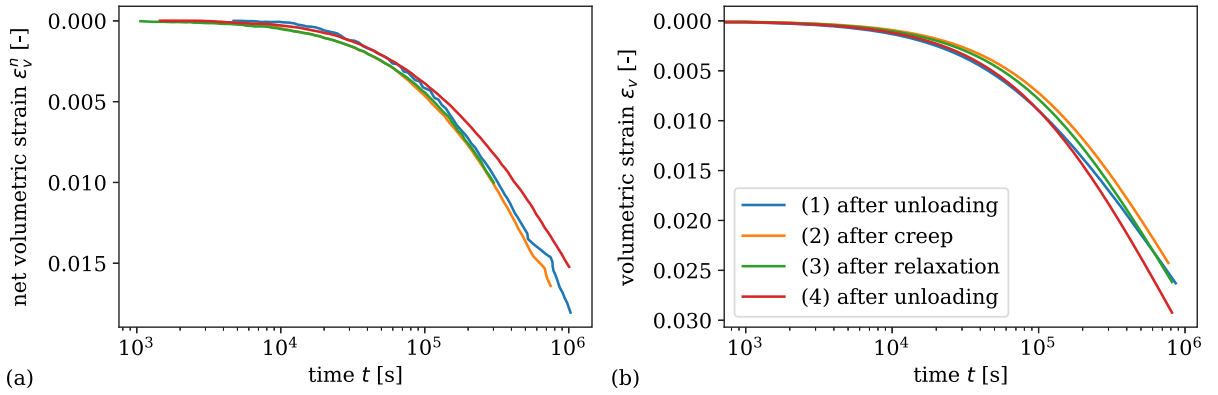


Fig. 16. Volumetric strain over time during the various creep stages of the comprehensive oedometer test in Fig. 15 from Niemunis and Krieg (1996) and Krieg (2000): (a) net experimental strain that excludes consolidation through fluid expulsion; and (b) simulated strain. The labels (1)–(4) refer to the respective creep stages shown on Fig. 15.

remarkable ability to explain the comprehensive experimental volumetric response, effectively capturing the constitutive response during the many different loading scenarios.

The evolution of the volumetric strain over time during the creep stages are plotted in Fig. 16 for both the simulation and the experiment. Qualitatively, the model's representation of creep over time aligns well with the experimental data. Notice that despite experimental suggestions for apparent independence of creep on preceding events (Krieg, 2000), the model indicates slight variations in the creep behaviour. In particular, it is important to note the generally lower strains in the experiments, since the data reported by the experimentalists reported a net volumetric strain ϵ_v^n whereby the strain contribution of consolidation through fluid expulsion was subtracted from the total volumetric strain ϵ_v . A more complete evaluation would require accounting for the water phase in the calculation of the thermodynamic pressure, and their gradients, as described by Chen et al. (2023). Nevertheless, the overall extent of the total extent of volumetric strains during creep stages is well predicted, as implied by the changes in solid fraction during those episodes in Fig. 15. Furthermore, while it could have been ideally interesting to assess the model's capacity of predicting stress variations with time during the relaxation phases, this was not supplied by the authors. It is, however, reassuring to notice that the model closely agrees in predicting the extent of stress reduction during the relaxation phase.

6.6. Potential application to rate and state in geological faults

While the preceding sections have explored rate-dependent phenomena in clay that are typically studied in soil mechanics—including creep, relaxation, and the effects of variable loading rates—rate-dependency of clay is also critically important for geophysicists in their study of clay-rich fault gouges. Clays retrieved from such geological features are often studied under biaxial conditions akin to simple shear, where they display similar forms of rate-dependent behaviours to those discussed in the previous section (Saffer and Marone, 2003; Carpenter et al., 2016). In geophysics, those features are typically explained through phenomenological rate and state models (Dieterich, 1979; Ruina, 1983; Heimissson et al., 2022; Pranger et al., 2022). To explore whether Terracotta can qualitatively capture the same ‘rate and state’ behaviour from first principles, we compare its predictions to experiments on actual fault materials. Unlike the previous evaluations of the model performance, this comparison is only conceptual. In particular, since Terracotta has been here formulated using triaxial invariants for simplicity, a proper description of its constitutive response under simple shear would require tensorial reformulation, which goes beyond the conceptual scope of this contribution. Despite this point, it is still instructive to adopt model simulations of undrained triaxial compression at high pressures and large strains as a proxy for qualitative comparisons against experimental responses under the sort of simple shear conditions used in geophysics.

With that in mind, Fig. 17 presents a detailed view of a shear test on illite shale by Saffer and Marone (2003), featuring step changes in shearing velocity. The depicted segment of the experiment occurs well beyond the point the mobilised friction $\mu = \tau/\sigma_n$ has achieved its steady/critical state value, and showcases the material response during two shearing rate step changes. The material's response in this geophysical study mirrors the so-called combined viscous behaviour in soil mechanics, which was discussed in Section 6.4. In particular, such a response shows either an abrupt increase or a decrease of the initial stress, followed by a relaxation towards a new steady state mobilised friction μ which is only dependent on the shearing rate.

Notice that due to the lack of experimental data and the qualitative purpose of the simulation, here the material parameters were calibrated superficially. Nevertheless, the model is clearly capable of capturing the pronounced peaks and subsequent relaxation in stress following step changes to both lower and higher loading strain rates from the preceding strain rates. Consistently, step increases in the rate induce less prominent stress peaks compared to decreases, a trend also noted in Fig. 13. As discussed previously, this behaviour is attributed to the dynamics of T_m which governs both viscous stress and plastic flow as detailed in Eqs. (29) and (32). Given the chosen linear dependencies of the transport coefficients on T_m as well as the applied strain rates, T_m exhibits a slower

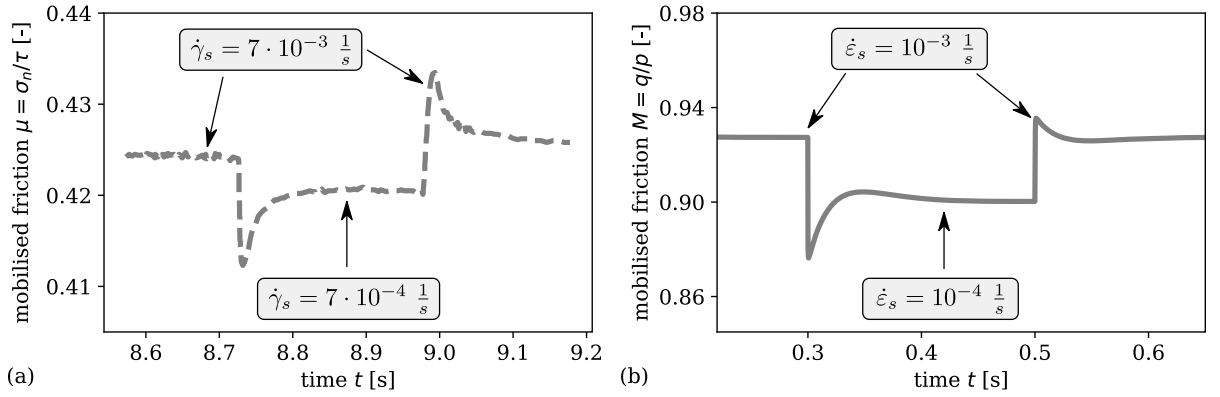


Fig. 17. Detailed view of rate and state responses of clay-rich geological faults to shearing strain rate steps: (a) during experimental simple shear test of illite shale by Saffer and Marone (2003) where $\dot{\gamma}_s$, σ_n and τ denote the simple shear strain rate, the normal stress, and the simple shear stress, respectively; and (b) during undrained triaxial compression of the Terracotta model.

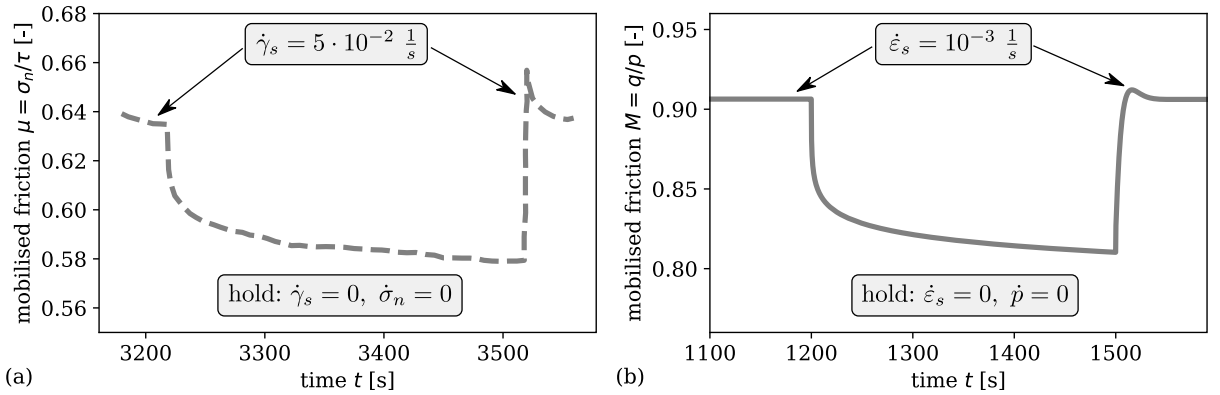


Fig. 18. Detailed view of a slide-hold-slide test response during: (a) experimental shear test on clay-bearing specimen from the Calaveras fault by Carpenter et al. (2016), and (b) simulated undrained triaxial compression test using the Terracotta model.

relaxation to its steady state compared to its rate of increase. Considering that the evaluation is based on only a first simple version of the Terracotta model, the evaluation clearly motivates further studies in geophysics. Future extensions may consider to vary the crucial coefficients $\bar{\alpha}$, $\bar{\gamma}$ and $\bar{\eta}$ with the elastic stress and density and non-linearly on T_m , rather than only linearly.

Another interesting way with which geophysicists characterise the rate-dependence of clay-rich fault materials is through slide-hold-slide tests designed to more closely mimic the seismic conditions controlling clay responses in geological faults. Associated with repeated earthquake failures (Brace, 1972), such tests describe periodic sequences of slip whereby the shear strain rate and normal stress are constant, and episodes of combined controlled creep-relaxation tests whereby the shear strain and normal stress are kept constant while the normal strain rate and the shear stress are allowed to change freely. In this type of experiment the ‘hold’ phase is typically introduced well after reaching the steady state value of the friction coefficient, as shown in Fig. 18 on a clay-bearing specimen from the Calaveras fault (Carpenter et al., 2016). Following the steady state μ , the specimen in the depicted experiment is subjected to a 300 s relaxation stage, before the shearing resumes using the initial velocity/shear rate. The material parameters for this simulation were only roughly adjusted from the previous one to study the capabilities of the model in this loading case. The hold phase in the analogous triaxial simulation is controlled by fixing the pressure and the shear strain while letting the volumetric strain and the deviatoric stress evolve. The simulation aligns qualitatively well with the experimental data during the relaxation phase, while further exhibiting a near-immediate rise in mobilised friction post-relaxation.

7. Conclusion

This paper introduces Terracotta, a simple and physically meaningful hydrodynamic constitutive model adept at capturing key rheological and mechanical phenomena in clays, such as asymptotic stress-density relation during monotonic isotropic compression, critical state upon monotonic shearing, rate and state response upon transient loading, as well as creep and relaxation. Developed through the thermodynamically rigorous hydrodynamic procedure (Landau and Lifshitz, 1980, 1987) and drawing inspiration from granular solid hydrodynamics (GSH) (Jiang and Liu, 2009), the development of the model employs classical state variables as well as introduces the meso-related temperature to capture the non-affine motion of meso-structures such as clay particles and aggregates.

One of the key steps in the development is the identification of the thermodynamic transport coefficients that couples the volumetric and shear plastic strain rates. In fact, this crucial dissipative coupling has not been identified throughout the development of past GSH models (Jiang and Liu, 2009, 2014). Instead, those models involved a rather more complex process. Firstly, in the two-stage irreversibility scheme of past GSH models the viscous contribution to the mechanical dissipation was split between the meso-related (granular) entropy production rate and the micro-related entropy production rate. This is unlike the simpler energy allocation specified in Eqs. (13) and (14), whereby the viscous part was attributed entirely to the meso-related (granular) entropy production rate. Secondly, in past GSH models the entire plastic strain rate was coupled to the micro-related part of the viscous (dissipative) stress, rather than coupling the volumetric and plastic strain rate fluxes as is done here for the first time. In this way, when compared with Terracotta, past GSH models involved quite a few more coefficients whose highly nonlinear dependence on the state variables has not been clarified and rather tuned through many more constants, the determination of which over large state regimes could not be well recovered. In summary, both the simpler treatment of two-stage irreversibility and the novel identification of the plastic transport coefficients are here found to be crucial.

The newly proposed method of calibrating the dependence of the plastic transport coefficients on the state variables is another novel step, and enables to recover experimental steady state observations, all without needing to recourse to classical plasticity concepts. This leads to a model that is both remarkably simple and firmly rooted in physical principles, bypassing intricate mathematical constructs like plastic multipliers. Despite its simplicity, as depicted in Fig. 1, the model robustly captures a plethora of clay behaviours. In particular, much thanks to the dependence of the plastic and viscous transport coefficients on the meso-related temperature, the model is readily capable of reproducing the diverse range of rate-dependent responses of clay under dynamically changing loading rates.

7.1. Perspectives

While Terracotta demonstrates excellent performance in capturing the qualitative behaviour of clay during monotonic and transient loading scenarios, it does not perfectly replicate stress variations across different loading rates. Under relatively low straining rates the meso-related temperature T_m quickly reaches its steady state value in Eq. (39). Under such conditions, the plastic strain rate varies linearly with the norm of the strain rate in Eq. (32) and the viscous stress quadratically with the strain rate in Eq. (29). This observation during steady state conditions highlight two potential aspects for future considerations.

Firstly, the linear dependence of Terracotta's plastic strain rate on the norm of the total strain rate under steady state is common to hypoplastic models (Kolymbas, 1991). Similar observation was made by Jiang and Liu (2009, 2016) in their granular solid hydrodynamic model for sand. Accordingly, it could be easily demonstrated that under relatively low strain rates, Terracotta can replicate the cyclical response of clay just as well as hypoplastic models do. However, just like hypoplasticity, Terracotta also overly 'ratchets', meaning it overpredicts the accumulation of irreversible deformation during cyclical loading. This observations present an excellent scope for future developments. In particular, in hypoplasticity this problem is often mitigated by introducing an additional tensorial state variable, called the inter-granular strain (Niemunis and Herle, 1997). Therefore, in order to address the same issue using hydrodynamics, one may consider to enhance Terracotta by adding such a state variable. However, unlike hypoplasticity, the hydrodynamic procedure places rigorous restrictions on the thermodynamic meaning of any constitutive assumption or new variables. Rather than seeing this as an obstacle, the casting of Terracotta within hydrodynamics actually opens up new ways to address the same problem. For example, one possibility would be to adopt a more subtle decomposition of the rate of entropy production between its meso-related and thermal parts. Such an approach was adopted by Jiang and Liu (2009, 2015), but would complicate the calibration of the transport coefficients. Therefore, perhaps a more promising approach would indeed be to add a new state variable. However, within the hydrodynamic rigour the thermodynamic nature of such a variable must be clarified, especially in whether it is an equilibrium (including elastic strain and density) or a non-equilibrium state variable. In the recent work of Einav and Liu (2023) such a non-equilibrium variable was employed to successfully explain hysteretic phenomena during cyclical drying and wetting of soils, similar to those happening during cyclic mechanical loading and unloading of soils.

Secondly, the quadratic dependence of Terracotta's viscous stress on the strain rate does not quantitatively align with experimental findings of a logarithmic or a power law relationship (Vaid et al., 1979; Graham et al., 1983; Leroueil et al., 1985). Reproducing such relationships would be required in order to represent clay responses to strain rate jumps of any order of magnitudes. In order to attain the observed relationship, the particular dependencies in Eqs. (31) and (35) of rheological coefficients $\bar{\alpha}$, $\bar{\gamma}$ and $\bar{\eta}$ on the meso-related temperature could be tuned while ensuring vanishing plastic strain rate and viscous stress on approach to thermodynamic equilibrium with $T_m \rightarrow 0$.

The novel approach in this paper also opens up new questions for perspective experimental research. In particular, it is important to recall the strategic omission through the use of $\bar{\beta} = 0$ of any possible cross-coupling between the volumetric and deviatoric viscous stresses and the corresponding strain rates ($\bar{\beta} \neq 0$). The reason for omitting this coefficient is the lack of data with which we could calibrate it. However, lack of data does not mean $\bar{\beta}$ is not important. According to the new theory, this coefficient would be required should a jump in deviatoric strain rate cause the overall pressure p to rapidly change, or conversely should a jump in volumetric strain rate cause the deviatoric stress q to rapidly vary. It follows that the determination of whether $\bar{\beta}$ is important, and hence its calibration, would require a new set of experiments that systematically vary those strain rates.

Furthermore, the development of Terracotta within the thermodynamically rigorous hydrodynamic procedure reveals a new mathematical structure in constitutive modelling. Accordingly, the proposed model paves the way for future extensions with which the broader spectrum of multi-physical phenomena could be replicated in a mathematically consistent manner. In particular, the behaviour of clay is profoundly influenced by environmental factors, such as temperature (Campanella and Mitchell, 1968) and

pore fluid chemistry (Sridharan and Rao, 1973), that are well within the scope that the hydrodynamic procedure can meaningfully represent. Notably, temperature is already intrinsically accounted for as a state parameter within our model, and thus can be seamlessly included in the internal energy density and the transport coefficients. The effects of changes in the chemical composition of the pore fluid in clay can be conceptually addressed, as was done by Chen et al. (2023) to model the chemical degradation of poroelastic rocks. Finally, thanks to the generality of hydrodynamics, it is anticipated that the capabilities of Terracotta could be expanded to capture fluid-solid transitions between liquid-like soft mud and plastic-like stiff clay responses, contingent on water content variations. At the limit of vanishing clay content, the corresponding clay suspension can be aptly described by the Navier–Stokes equations, which are naturally recovered using the hydrodynamic procedure (Landau and Lifshitz, 1987). To encompass the fluid phase, the model could automatically benefit from recent developments on multi-phase soil media by Jiang et al. (2017) and Einav and Liu (2018, 2023). Successfully capturing the solid–fluid transition for clay holds the potential to model phenomena like rainfall-induced liquefaction and subsequent mudflows. As we continue to develop and refine Terracotta, the simplicity of its equations and its rigorous foundation provide new means with which the community could further the understanding and prediction of the unique constitutive behaviour of clay in a myriad of environmental conditions.

CRedit authorship contribution statement

Max Wiebicke: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Itai Einav:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The implementation of Terracotta accompanying this manuscript is available at github.com/MaxWiebicke/Terracotta-pq.

Acknowledgements

The authors would like to thank Mario Liu for insightful discussions regarding his seminal work on hydrodynamics. Special thanks are also extended to Yanni Chen, François Guillard, Filippo Masi and David Riley, for valuable discussions about the development of Terracotta. The research leading to these results has received funding from the German Research Foundation (DFG) through grant No. 457695076, and the Australian Research Council (ARC) through DP220101164.

Appendix A. Effective and elastic stresses

The decomposition of the total stress as given in Eq. (4) is a fundamental result of the hydrodynamic procedure, for any material, not just clay. Here, we demonstrate the strong connection of this result to the effective stress principle in soil mechanics in the specific context of Terracotta. First, from Eq. (4) the elastic stress from hydrodynamics is given as:

$$\sigma_{ij}^e = \sigma_{ij} - \sigma_{ij}^d - p^T \delta_{ij}. \quad (\text{A.1})$$

Next, using the formulation of Terracotta in the paper we notice from Eq. (37) that the thermodynamic pressure p^T has both an elastic p_e^T and a meso-related p_m^T term. However, the corresponding derivation considered clays as if they reside in a vacuum, or at least not affected by air pressure. In the presence of fluids, it was shown by Jiang et al. (2017) that the hydrodynamic procedure leads to additional fluid pressure term P , so that:

$$p^T = p_m^T + p_e^T + P. \quad (\text{A.2})$$

When Terracotta is saturated in water, the fluid pressure $P = u_w$, when saturated in air we get $P = u_a$, and when partially saturated through air–water mixture $P = u_a + \chi(u_a - u_w)$. Here, u_a , u_w and χ are the air pressure, water pressure and Bishop's coefficient (Bishop, 1959), respectively. The structure of χ was derived in Jiang et al. (2017) through the hydrodynamic procedure in exact form in terms of soil-fluid retention characteristics.

Next, we notice that the elastic contribution to the thermodynamic pressure p_e^T is generally negligible compared to the elastic stress, so Terracotta yields similar results whether it is considered or not, as shown in Appendix B. Further notice that in Terracotta, under slow shear rates both the viscous stress and the meso-related thermodynamic pressure vanish too. Accordingly, we find the elastic stress is fully consistent with the effective stress from soil mechanics:

$$\sigma'_{ij} \equiv \sigma_{ij}^e = \sigma_{ij} - P \delta_{ij}, \quad (\text{A.3})$$

where the equivalency between the effective stress and elastic stress was first identified by Jiang et al. (2017).

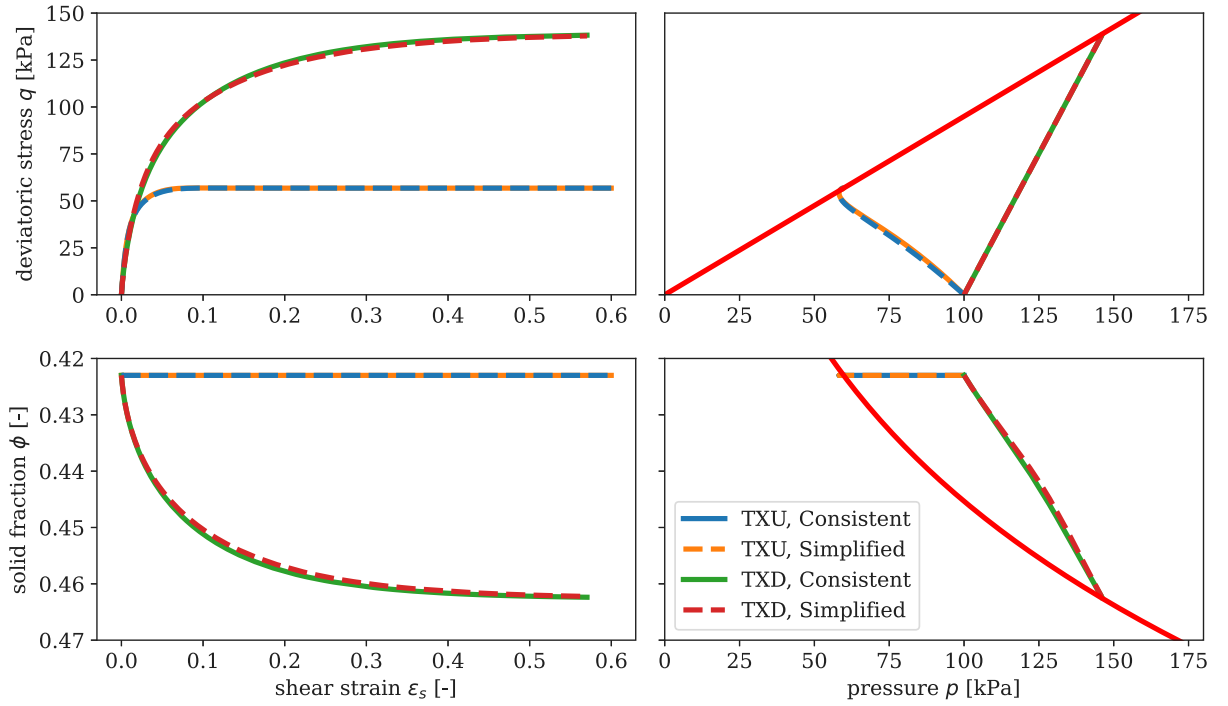


Fig. B.19. Constitutive response of the simplified and the fully consistent Terracotta models of Karlsruhe kaolin using constants in Table 1 during drained (TXD) and undrained (TXU) triaxial compression tests.

Appendix B. Fully consistent model formulation

The Terracotta model illustrated in Fig. 1 follows a simplified approximated form by neglecting the elastic term p_e^T in the thermodynamic pressure p^T relative to the elastic pressure p^e within the total pressure p , as elaborated in Section 4.6. Here this approximation is assessed and demonstrated to be appropriate. To this end, the fully consistent model follows the complete solution for p^T given by Eq. (37). Consequently, rather than using the approximated solution for the total pressure p shown in Eq. (38), in the fully consistent model:

$$p = p^e + p^d + p_e^T + p_m^T. \quad (\text{B.1})$$

The viscous transport coefficients in the consistent model remain identical thanks to their definition in Eq. (31). However, adopting the equation above for p requires to recalibrate the plastic transport coefficients, as follows from Section 5. This is necessary since those coefficients were identified from experimental observations on the steady state response of clay in isotropic and triaxial compression, where p varies.

As before, considering very small loading rates, the viscous pressure and meso-related temperature are neglected, from which $p \cong p^e + p_e^T$. Using this relation rather than $p \cong p^e$ from before, the plastic transport coefficients of the fully consistent formulation of Terracotta become:

$$\bar{a} = \sqrt{\frac{\eta}{\alpha}} \frac{1}{p_c(\phi) - p_e^T}, \quad (\text{B.2})$$

$$\bar{b} = -\bar{a} \frac{p^e q^e}{M^2 (p^e + p_e^T)^2}, \quad (\text{B.3})$$

$$\bar{c} = \sqrt{\frac{\eta}{\gamma}} \frac{1}{M p_c(\phi)} + \bar{a} \frac{(\omega p_c(\phi) - p_e^T)^2}{(M \omega p_c(\phi))^2}. \quad (\text{B.4})$$

In order to compare the simplified and the fully consistent version of Terracotta, Fig. B.19 presents simulations of one undrained and one drained triaxial compression tests for each model. Importantly, for exactly the same material constants for Karlsruhe kaolin in Table 1 the numerical results obtained by both models are practically unaffected, thus justifying the use of the approximated model in Fig. 1.

Appendix C. Notation

The following table provides the physical dimensions of all symbols employed in this paper. In this context, L denotes the length, T the time, M the mass, and θ the thermal temperature.

Symbol	Name/Description	Dimension
$\bar{a}$	Volumetric plastic transport coefficient	$LT M^{-1} \theta^{-1}$
$\bar{b}$	Coupled volumetric-deviatoric plastic transport coefficient	$LT M^{-1} \theta^{-1}$
$\bar{c}$	Deviatoric plastic transport coefficient	$LT M^{-1} \theta^{-1}$
f_i	Thermal entropy current	$MT^{-3} \theta^{-1}$
f_i^m	Meso-related entropy current	$MT^{-3} \theta^{-1}$
G	Instantaneous shear modulus	$ML^{-1} T^{-2}$
$\bar{G}$	Intrinsic shear modulus	$ML^{-1} T^{-2}$
K	Instantaneous bulk modulus	$ML^{-1} T^{-2}$
$\bar{K}$	Intrinsic bulk modulus	$ML^{-1} T^{-2}$
M	Critical stress ratio	–
M_e	Critical stress ratio regarding elastic instability	–
M_o	Critical stress ratio regarding Onsagers reciprocal relations	–
p	Total pressure	$ML^{-1} T^{-2}$
$p_c(\phi)$	Isotropic compression line (ICL)	$ML^{-1} T^{-2}$
p_l	Constant defining the ICL	$ML^{-1} T^{-2}$
p^d	Viscous pressure	$ML^{-1} T^{-2}$
p^e	Elastic pressure	$ML^{-1} T^{-2}$
p^T	Thermodynamic pressure	$ML^{-1} T^{-2}$
p_m^T	meso-related contribution to p^T	$ML^{-1} T^{-2}$
p_e^T	elastic contribution to p^T	$ML^{-1} T^{-2}$
q	Deviatoric total stress	$ML^{-1} T^{-2}$
q^d	Deviatoric viscous stress	$ML^{-1} T^{-2}$
q^e	Deviatoric elastic stress	$ML^{-1} T^{-2}$
R	Thermal entropy production	$ML^{-1} T^{-3}$
R_m	Meso-related entropy production	$ML^{-1} T^{-3}$
s	Thermal entropy	$ML^{-1} T^{-2} \theta^{-1}$
s_m	Meso-related entropy	$ML^{-1} T^{-2} \theta^{-1}$
T	Thermal temperature	θ
T_m	Meso-related temperature	θ
u	Internal energy density	$ML^{-1} T^{-2}$
u^e	Internal elastic energy density	$ML^{-1} T^{-2}$
u^m	Internal meso-related energy	$ML^{-1} T^{-2}$
v_i	Velocity vector	LT^{-1}
X_{ijkl}^d	Viscous transport coefficients	$ML^{-1} T^{-1} \theta^{-1}$
X_{ijkl}^p	Plastic transport coefficients	$LT M^{-1} \theta^{-1}$
α	Constant parameter setting the volumetric viscous stress	$T \theta$
$\bar{\alpha}$	Volumetric viscous transport coefficient	$ML^{-1} T^{-1} \theta^{-1}$
$\bar{\beta}$	Coupled volumetric-deviatoric viscous transport coefficient	$ML^{-1} T^{-1} \theta^{-1}$
γ	Constant parameter setting the deviatoric viscous stress	$T \theta$
$\bar{\gamma}$	Deviatoric viscous transport coefficient	$ML^{-1} T^{-1} \theta^{-1}$
Γ	Fixed coefficient that sets the unit of the meso-related temperature with $\Gamma = 1 \text{ K}^2/\text{kPa}$	$M^{-1} LT^2 \theta^2$
$\bar{\eta}$	Energy sink coefficient	$ML^{-1} T^{-3} \theta^{-2}$
η	Constant parameter specifying the energy sink	$T^{-1} \theta^{-1}$
δ_{ij}	Kronecker delta tensor	–
ϵ_{ij}^e	Elastic strain tensor	–
ϵ_v^e	Volumetric elastic strain	–
ϵ_s^e	Deviatoric elastic strain	–
$\dot{\epsilon}_{ij}$	Total strain rate tensor	T^{-1}
$\dot{\epsilon}_{ij}^e$	Elastic strain rate tensor	T^{-1}
$\dot{\epsilon}_{ij}^p$	Plastic strain rate tensor	T^{-1}
μ	Chemical potential	$L^2 T^{-2}$
ϕ	Solid fraction	–
ϕ_1	Solid fraction of the ICL at $p = 1 \text{ kPa}$	–

λ	Slope of the ICL	–
ρ	Bulk density	ML^{-3}
ρ_s	Solid density	ML^{-3}
ρ_s^*	Unstressed solid density	ML^{-3}
σ_{ij}^d	Total stress tensor	$ML^{-1}T^{-2}$
σ_{ij}^v	Viscous stress tensor	$ML^{-1}T^{-2}$
σ_{ij}^e	Elastic stress tensor	$ML^{-1}T^{-2}$
ω	Position of the critical state line	–
$\dot{\square}$	Rate of variable $\square$	$\square T^{-1}$
$\nabla_i \square$	Gradient of variable $\square$	$\square L^{-1}$
$\nabla_i \square_i$	Divergence of variable $\square_i$	$\square L^{-1}$

References

- Adachi, T., Oka, F., 1982. Constitutive equations for normally consolidated clay based on elasto-viscoplasticity. *Soils Found.* 22 (4), 57–70. http://dx.doi.org/10.3208/sandf1972.22.4_57.
- Alaei, E., Marks, B., Einav, I., 2021. A hydrodynamic-plastic formulation for modelling sand using a minimal set of parameters. *J. Mech. Phys. Solids* 151, 104388. <http://dx.doi.org/10.1016/j.jmps.2021.104388>.
- Alaei, E., Marks, B., Einav, I., 2022. A five-parameter hydrodynamic-plastic model for crushable sand. *Int. J. Solids Struct.* 254, 111914. <http://dx.doi.org/10.1016/j.ijsolstr.2022.111914>.
- Amorosi, A., Rollo, F., Dafalias, Y.F., 2021. Relating elastic and plastic fabric anisotropy of clays. *Géotechnique* 71 (7), 583–593. <http://dx.doi.org/10.1680/jgeot.19.P.134>.
- Bishop, A.W., 1959. The principle of effective stress. *Tek. Ukebl.* 39, 859–863.
- Borja, R., Kavazanjian, E., 1985. A constitutive model for the stress–strain–time behaviour of ‘wet’ clays. *Géotechnique* 35 (3), 283–298. <http://dx.doi.org/10.1680/geot.1985.35.3.283>.
- Brace, W., 1972. Laboratory studies of stick-slip and their application to earthquakes. *Tectonophysics* 14 (3–4), 189–200. [http://dx.doi.org/10.1016/0040-1951\(72\)90068-6](http://dx.doi.org/10.1016/0040-1951(72)90068-6).
- Buisman, A., 1936. Results of long duration settlement tests. In: *Proc. 1st ICSMFE. Vol. 1, Cambridge*, pp. 103–107.
- Burland, J., 1990. On the compressibility and shear strength of natural clays. *Géotechnique* 40 (3), 329–378. <http://dx.doi.org/10.1680/geot.1990.40.3.329>.
- Butterfield, R., 1979. A natural compression law for soils (an advance on e – $\log p$). *Géotechnique* 29 (4), 469–480. <http://dx.doi.org/10.1680/geot.1979.29.4.469>.
- Campanella, R.G., Mitchell, J.K., 1968. Influence of temperature variations on soil behavior. *J. Soil Mech. Found. Div.* 94 (3), 709–734. <http://dx.doi.org/10.1061/JSEFAQ.0001136>.
- Carlton, B., Pestana, J., 2012. Small strain shear modulus of high and low plasticity clays and silts. In: *Proc., 15th World Conf. on Earthquake Engineering. Tokyo: International Association for Earthquake Engineering*.
- Carpenter, B.M., Ikari, M.J., Marone, C., 2016. Laboratory observations of time-dependent frictional strengthening and stress relaxation in natural and synthetic fault gouges. *J. Geophys. Res.: Solid Earth* 121 (2), 1183–1201. <http://dx.doi.org/10.1002/2015JB012136>.
- Chen, Y., Guillard, F., Einav, I., 2023. A hydrodynamic model for chemo-mechanics of poroelastic materials. *Géotechnique* <http://dx.doi.org/10.1680/jgeot.22.00260>.
- Cotecchia, F., Chandler, R., 2000. A general framework for the mechanical behaviour of clays. *Géotechnique* 50 (4), 431–447. <http://dx.doi.org/10.1680/geot.2000.50.4.431>.
- Cotecchia, F., Guglielmi, S., Cafaro, F., Gens, A., 2019. Characterisation of the multi-scale fabric features of high plasticity clays. *Géotech. Lett.* 9 (4), 361–368. <http://dx.doi.org/10.1680/jgele.18.00230>.
- De Groot, S., Mazur, P., 1954. Extension of Onsager’s theory of reciprocal relations. I. *Phys. Rev.* 94 (2), 218. <http://dx.doi.org/10.1103/PhysRev.94.218>.
- Delage, P., 2010. A microstructure approach to the sensitivity and compressibility of some eastern Canada sensitive clays. *Géotechnique* 60 (5), 353–368. <http://dx.doi.org/10.1680/geot.2010.60.5.353>.
- Delage, P., Lefebvre, G., 1984. Study of the structure of a sensitive champlain clay and of its evolution during consolidation. *Can. Geotech. J.* 21 (6), 209. <http://dx.doi.org/10.1139/t84-003>.
- Dieterich, J.H., 1979. Modeling of rock friction: 2. Simulation of preseismic slip. *J. Geophys. Res.: Solid Earth* 84 (B5), 2169–2175. <http://dx.doi.org/10.1029/JB084iB05p02169>.
- Duque, J., Roháč, J., Mašin, D., Najser, J., 2022. Experimental investigation on Malaysian kaolin under monotonic and cyclic loading: inspection of undrained Miner’s rule and drained cyclic preloading. *Acta Geotech.* 17 (11), 4953–4975. <http://dx.doi.org/10.1007/s11440-022-01643-0>.
- Einav, I., Liu, M., 2018. Hydrodynamic derivation of the work input to fully and partially saturated soils. *J. Mech. Phys. Solids* 110, 205–217. <http://dx.doi.org/10.1016/j.jmps.2017.10.004>.
- Einav, I., Liu, M., 2023. Hydrodynamics of non-equilibrium soil water retention. *Water Resour. Res.* 59 (1), <http://dx.doi.org/10.1029/2022WR033409>.
- Fuentes, W., Tafil, M., Triantafyllidis, T., 2018. An ISA-plasticity-based model for viscous and non-viscous clays. *Acta Geotech.* 13, 367–386. <http://dx.doi.org/10.1007/s11440-017-0548-y>.
- Furukawa, Y., Watkins, J.L., Kim, J., Curry, K.J., Bennett, R.H., 2009. Aggregation of montmorillonite and organic matter in aqueous media containing artificial seawater. *Geochem. Trans.* 10, 1–11. <http://dx.doi.org/10.1186/1467-4866-10-2>.
- Gasparre, A., 2005. *Advanced Laboratory Characterisation of London Clay* (Ph.D. thesis). Imperial College, London.
- Gasparre, A., Nishimura, S., Coop, M.R., Jardine, R.J., 2007. The influence of structure on the behaviour of London Clay. *Géotechnique* 57 (1), 19–31. <http://dx.doi.org/10.1680/geot.2007.57.1.19>.
- Goldhirsch, I., 2008. Introduction to granular temperature. *Powder Technol.* 182 (2), 130–136. <http://dx.doi.org/10.1016/j.powtec.2007.12.002>.
- Graham, J., Crooks, J., Bell, A.L., 1983. Time effects on the stress–strain behaviour of natural soft clays. *Géotechnique* 33 (3), 327–340. <http://dx.doi.org/10.1680/geot.1983.33.3.327>.
- Grim, R.E., 1962. *Applied Clay Mineralogy*. McGraw-Hill.
- Gudehus, G., Amorosi, A., Gens, A., Herle, I., Kolymbas, D., Mašin, D., Muir Wood, D., Niemunis, A., Nova, R., Pastor, M., et al., 2008. The soilmodels. info project. *Int. J. Numer. Anal. Methods Geomech.* 32 (12), 1571–1572. <http://dx.doi.org/10.1002/nag.675>.

- Guglielmi, S., Cotecchia, F., Cafaro, F., Gens, A., 2022. Analysis of the micro to macro response of clays to compression. *Géotechnique* <http://dx.doi.org/10.1680/jgeot.21.00233>.
- Gupta, V., Hampton, M.A., Stokes, J.R., Nguyen, A.V., Miller, J.D., 2011. Particle interactions in kaolinite suspensions and corresponding aggregate structures. *J. Colloid Interface Sci.* 359 (1), 95–103. <http://dx.doi.org/10.1016/j.jcis.2011.03.043>.
- Hardin, B.O., Black, W.L., 1968. Vibration modulus of normally consolidated clay. *J. Soil Mech. Found. Div.* 94 (2), 353–369. <http://dx.doi.org/10.1061/JSFEAQ.000110>.
- Hashiguchi, K., Ueno, M., 1977. Elasto-plastic constitutive laws for granular materials, constitutive equations for soils. In: *Constitutive Equations for Soils, Proc. Specialty Session 9, 9th ICSMFE, Tokyo*. pp. 73–82.
- Heimisson, E.R., Smith, J.D., Avouac, J.-P., Bourne, S.J., 2022. Coulomb threshold rate-and-state model for fault reactivation: application to induced seismicity at groningen. *Geophys. J. Int.* 228 (3), 2061–2072. <http://dx.doi.org/10.1093/gji/ggab467>.
- Henkel, D., 1956. The effect of overconsolidation on the behaviour of clays during shear. *Geotechnique* 6 (4), 139–150. <http://dx.doi.org/10.1680/geot.1956.6.4.139>.
- Hicher, P.Y., Wahyudi, H., Tessier, D., 2000. Microstructural analysis of inherent and induced anisotropy in clay. *Mech. Cohes.-Frict. Mater.* 5 (5), 341–371. [http://dx.doi.org/10.1002/1099-1484\(200007\)5:5<341::AID-CFM99>3.0.CO;2-C](http://dx.doi.org/10.1002/1099-1484(200007)5:5<341::AID-CFM99>3.0.CO;2-C).
- Houlsby, G., Amorosi, A., Rollo, F., 2019. Non-linear anisotropic hyperelasticity for granular materials. *Comput. Geotech.* 115, 103167. <http://dx.doi.org/10.1016/j.compgeo.2019.103167>.
- Jamiolkowski, M., Lancellotta, R., Lo Presti, D., 1995. Remarks on the stiffness at small strains of six Italian clays. In: *Pre-Failure Deformation of Geomaterials. Proceedings of the International Symposium, 12-14 September 1994, Sapporo, Japan*. 2 Vols.
- Jiang, Y., Einav, I., Liu, M., 2017. A thermodynamic treatment of partially saturated soils revealing the structure of effective stress. *J. Mech. Phys. Solids* 100, 131–146. <http://dx.doi.org/10.1016/j.jmps.2016.11.018>.
- Jiang, Y., Liu, M., 2003. Granular elasticity without the Coulomb condition. *Phys. Rev. Lett.* 91 (14), 144301. <http://dx.doi.org/10.1103/PhysRevLett.91.144301>.
- Jiang, Y., Liu, M., 2009. Granular solid hydrodynamics. *Granul. Matter* 11, 139–156. <http://dx.doi.org/10.1007/s10035-009-0137-3>.
- Jiang, Y., Liu, M., 2014. Granular solid hydrodynamics (GSH): a broad-ranged macroscopic theory of granular media. *Acta Mech.* 225, 2363–2384. <http://dx.doi.org/10.1007/s00707-014-1131-3>.
- Jiang, Y., Liu, M., 2015. Applying GSH to a wide range of experiments in granular media. *Eur. Phys. J. E* 38, 1–27. <http://dx.doi.org/10.1140/epje/i2015-15015-6>.
- Jiang, Y., Liu, M., 2016. Similarities between GSH, hypoplasticity and KCR. *Acta Geotech.* 11, 519–537. <http://dx.doi.org/10.1007/s11440-016-0461-9>.
- Kolymbas, D., 1991. An outline of hypoplasticity. *Arch. Appl. Mech.* 61 (3), 143–151. <http://dx.doi.org/10.1007/BF00788048>.
- Kolymbas, D., 2000. The misery of constitutive modelling. In: *Constitutive Modelling of Granular Materials*. Springer, pp. 11–24. http://dx.doi.org/10.1007/978-3-642-57018-6_1.
- Krieg, S., 2000. *Viskoses Bodenverhalten von Mudden, Seeton und Klei* (Ph.D. thesis). Institut für Boden- und Felsmechanik der Universität Fridericiana Karlsruhe, Heft 150.
- Lacerda, W.A., Houston, W.N., 1973. Stress relaxation in soils. In: *Proc. 8th ICSMFE, Moscow, 1973*. pp. 221–227.
- Landau, L.D., Lifshitz, E.M., 1980. *Statistical Physics*. Butterworth-Heinemann, <http://dx.doi.org/10.1016/C2009-0-24487-4>.
- Landau, L.D., Lifshitz, E.M., 1987. *Fluid Mechanics*. Butterworth, <http://dx.doi.org/10.1016/C2013-0-03799-1>.
- Le, T.M., Fatahi, B., Khabbaz, H., 2012. Viscous behaviour of soft clay and inducing factors. *Geotech. Geol. Eng.* 30, 1069–1083. <http://dx.doi.org/10.1007/s10706-012-9535-0>.
- Leroueil, S., Kabbaj, M., Tavenas, F., Bouchard, R., 1985. Stress–strain–strain rate relation for the compressibility of sensitive natural clays. *Géotechnique* 35 (2), 159–180. <http://dx.doi.org/10.1680/geot.1985.35.2.159>.
- Mansa, R., Ngassa Piegang, G.B., Detellier, C., 2017. Kaolinite aggregation in book-like structures from non-aqueous media. *Clays Clay Miner.* 65 (3), 193–205. <http://dx.doi.org/10.1346/CCMN.2017.064059>.
- Maranic, Z., Guillard, F., Baker, J., Einav, I., Marks, B., 2021. A granular thermometer. *Granul. Matter* 23, 1–15. <http://dx.doi.org/10.1007/s10035-020-01081-3>.
- Missana, T., Alonso, U., Fernández, A.M., García-Gutiérrez, M., 2018. Analysis of the stability behaviour of colloids obtained from different smectite clays. *Appl. Geochem.* 92, 180–187. <http://dx.doi.org/10.1016/j.apgeochem.2018.03.010>.
- Mitchell, J.K., Soga, K., 2005. *Fundamentals of Soil Behavior*, vol. 3, John Wiley & Sons, New York.
- Nguyen, G.D., Einav, I., 2009. The energetics of cataclasis based on breakage mechanics. *Pure Appl. Geophys.* 166, 1693–1724. <http://dx.doi.org/10.1007/s00024-009-0518-x>.
- Niemunis, A., 2003. *Extended Hypoplastic Models for Soils*. In: *Monografia*, vol. 34, Ruhr-University Bochum.
- Niemunis, A., Herle, I., 1997. Hypoplastic model for cohesionless soils with elastic strain range. *Mech. Cohes.-Frict. Mater.* 2 (4), 279–299. [http://dx.doi.org/10.1002/\(SICI\)1099-1484\(199710\)2:4<279::AID-CFM29>3.0.CO;2-8](http://dx.doi.org/10.1002/(SICI)1099-1484(199710)2:4<279::AID-CFM29>3.0.CO;2-8).
- Niemunis, A., Krieg, S., 1996. Viscous behaviour of soil under oedometric conditions. *Can. Geotech. J.* 33 (1), 159–168. <http://dx.doi.org/10.1139/t96-031>.
- Onsager, L., 1931. Reciprocal relations in irreversible processes. I. *Phys. Rev.* 37 (4), 405. <http://dx.doi.org/10.1103/PhysRev.37.405>.
- Parry, R., 1960. Triaxial compression and extension tests on remoulded saturated clay. *Géotechnique* 10 (4), 166–180. <http://dx.doi.org/10.1680/geot.1960.10.4.166>.
- Pranger, C., Sanan, P., May, D.A., Le Pourhiet, L., Gabriel, A.-A., 2022. Rate and state friction as a spatially regularized transient viscous flow law. *J. Geophys. Res.: Solid Earth* 127 (6), <http://dx.doi.org/10.1029/2021JB023511>.
- Psyrillos, A., Howe, J., Manning, D., Burley, S., 1999. Geological controls on kaolin particle shape and consequences for mineral processing. *Clay Miner.* 34 (1), 193–208. <http://dx.doi.org/10.1180/000985599546037>.
- Riley, D.M., 2023. *Compaction Dynamics in Porous Media: a Combined Experimental, Numerical, and Analytical Approach* (Ph.D. thesis). The University of Sydney.
- Riley, D.M., Einav, I., Guillard, F., 2023a. A consistent derivation of soil stiffness from elastic wave speeds. <http://dx.doi.org/10.48550/arXiv.2312.01666>, [arXiv:2312.01666](https://arxiv.org/abs/2312.01666).
- Riley, D.M., Einav, I., Guillard, F., 2023b. A constitutive model for porous media with recurring stress drops: From snow to foams and cereals. *Int. J. Solids Struct.* 262, 112044. <http://dx.doi.org/10.1016/j.ijsolstr.2022.112044>.
- Roscoe, K., Burland, J.B., 1968. On the generalized stress-strain behaviour of wet clay. *Eng. Plast.* 535–609.
- Rubin, M.B., Einav, I., 2011. A large deformation breakage model of granular materials including porosity and inelastic distortional deformation rate. *Internat. J. Engng. Sci.* 49 (10), 1151–1169. <http://dx.doi.org/10.1016/j.ijengsci.2011.05.002>.
- Ruina, A., 1983. Slip instability and state variable friction laws. *J. Geophys. Res.: Solid Earth* 88 (B12), 10359–10370. <http://dx.doi.org/10.1029/JB088iB12p10359>.
- Saffer, D.M., Marone, C., 2003. Comparison of smectite-and illite-rich gouge frictional properties: application to the updip limit of the seismogenic zone along subduction megathrusts. *Earth Planet. Sci. Lett.* 215 (1–2), 219–235. [http://dx.doi.org/10.1016/S0012-821X\(03\)00424-2](http://dx.doi.org/10.1016/S0012-821X(03)00424-2).
- Santamarina, J.C., Klein, A., Fam, M.A., 2001. *Soils and Waves: Particulate Materials Behaviour, Characterisation and Process Monitoring*. John Wiley & Sons, Inc., <http://dx.doi.org/10.1007/BF02987719>.
- Schofield, A.N., Wroth, P., 1968. *Critical State Soil Mechanics*, vol. 310, McGraw-Hill, London.
- Skempton, A.W., Jones, O., 1944. Notes on the compressibility of clays. *Q. J. Geol. Soc.* 100 (1–4), 119–135. <http://dx.doi.org/10.1144/GSL.JGS.1944.100.01-04.08>.

- Sorensen, K., Baudet, B., Simpson, B., 2007. Influence of structure on the time-dependent behaviour of a stiff sedimentary clay. *Géotechnique* 57 (1), 113–124. <http://dx.doi.org/10.1680/geot.2007.57.1.113>.
- Sridharan, A., Rao, G.V., 1973. Mechanisms controlling volume change of saturated clays and the role of the effective stress concept. *Géotechnique* 23 (3), 359–382. <http://dx.doi.org/10.1680/geot.1973.23.3.359>.
- Šuklje, L., 1969. *Rheological Aspects of Soil Mechanics*. Wiley-Interscience London.
- Tafili, M., Fuentes, W., Triantafyllidis, T., 2020. A comparative study of different model families for the constitutive simulation of viscous clays. *Int. J. Numer. Anal. Methods Geomech.* 44 (5), 633–667. <http://dx.doi.org/10.1002/nag.3024>.
- Tan, X., Liu, F., Hu, L., Reed, A.H., Furukawa, Y., Zhang, G., 2017. Evaluation of the particle sizes of four clay minerals. *Appl. Clay Sci.* 135, 313–324. <http://dx.doi.org/10.1016/j.clay.2016.10.012>.
- Tarantino, A., De Col, E., 2008. Compaction behaviour of clay. *Géotechnique* 58 (3), 199–213. <http://dx.doi.org/10.1680/geot.2008.58.3.199>.
- Tatsuoka, F., Di Benedetto, H., Enomoto, T., Kawabe, S., Kongkitkul, W., 2008. Various viscosity types of geomaterials in shear and their mathematical expression. *Soils Found.* 48 (1), 41–60. <http://dx.doi.org/10.3208/sandf.48.41>.
- Tatsuoka, F., Ishihara, M., Di, B.H., Kuwano, R., 2002. Time-dependent shear deformation characteristics of geomaterials and their simulation. *Soils Found.* 42 (2), 103–129. <http://dx.doi.org/10.3208/sandf.42.2.103>.
- Terzaghi, K., 1925. *Erdbaumechanik auf Bodenphysikalischer Grundlage*. Deuticke.
- Terzaghi, K., Peck, R.B., 1948. *Soil mechanics*. In: *Engineering Practice*. John Wiley and Sons, Inc., New York.
- Tombácz, E., Csanaky, C., Illés, E., 2001. Polydisperse fractal aggregate formation in clay mineral and iron oxide suspensions, pH and ionic strength dependence. *Colloid Polym. Sci.* 279, 484–492. <http://dx.doi.org/10.1007/s003960100480>.
- Vaid, Y., Robertson, P., Campanella, R., 1979. Strain rate behaviour of saint-jean-vianney clay. *Can. Geotech. J.* 16 (1), 34–42. <http://dx.doi.org/10.1139/t79-004>.
- Virtanen, P., Gommers, R., Oliphant, T.E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S.J., Brett, M., Wilson, J., Millman, K.J., Mayorov, N., Nelson, A.R.J., Jones, E., Kern, R., Larson, E., Carey, C.J., Polat, İ., Feng, Y., Moore, E.W., VanderPlas, J., Laxalde, D., Perktold, J., Cimrman, R., Henriksen, I., Quintero, E.A., Harris, C.R., Archibald, A.M., Ribeiro, A.H., Pedregosa, F., van Mulbregt, P., SciPy 1.0 Contributors, 2020. SciPy 1.0: Fundamental algorithms for scientific computing in python. *Nat. Methods* 17, 261–272. <http://dx.doi.org/10.1038/s41592-019-0686-2>.
- Walton, O.R., Braun, R.L., 1986. Viscosity, granular-temperature, and stress calculations for shearing assemblies of inelastic, frictional disks. *J. Rheol.* 30 (5), 949–980. <http://dx.doi.org/10.1122/1.549893>.
- Wichtmann, T., Triantafyllidis, T., 2018. Monotonic and cyclic tests on kaolin: a database for the development, calibration and verification of constitutive models for cohesive soils with focus to cyclic loading. *Acta Geotech.* 13 (5), 1103–1128. <http://dx.doi.org/10.1007/s11440-017-0588-3>.
- Wood, D.M., 1990. *Soil Behaviour and Critical State Soil Mechanics*. Cambridge University Press, <http://dx.doi.org/10.1017/CBO9781139878272>.
- Yin, J.-H., Graham, J., 1999. Elastic viscoplastic modelling of the time-dependent stress-strain behaviour of soils. *Can. Geotech. J.* 36 (4), 736–745. <http://dx.doi.org/10.1139/t99-042>.
- Yuan, Y., Whittle, A.J., 2013. Examination on time-dependent soil models in one-dimensional consolidation. In: Yang, Q., Zhang, J.-M., Zheng, H., Yao, Y. (Eds.), *Constitutive Modeling of Geomaterials*. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 159–166. http://dx.doi.org/10.1007/978-3-642-32814-5_18.
- Yuan, Y., Whittle, A.J., 2018. A novel elasto-viscoplastic formulation for compression behaviour of clays. *Géotechnique* 68 (12), 1044–1055. <http://dx.doi.org/10.1680/jgeot.16.P.276>.
- Yuan, Y., Whittle, A.J., 2021. Formulation of a new elastoviscoplastic model for time-dependent behavior of clay. *Int. J. Numer. Anal. Methods Geomech.* 45 (6), 843–864. <http://dx.doi.org/10.1002/nag.3174>.
- Zhang, Z.-C., Cheng, X.-H., 2015. A thermodynamic constitutive model for undrained monotonic and cyclic shear behavior of saturated soils. *Acta Geotech.* 10, 781–796. <http://dx.doi.org/10.1007/s11440-015-0389-5>.
- Zhang, Z.-C., Cheng, X.-H., 2016. A thermo-mechanical coupled constitutive model for clay based on extended granular solid hydrodynamics. *Comput. Geotech.* 80, 373–382. doi:j.comptech.2016.05.010.